



NATIONAL TECHNICAL UNIVERSITY OF ATHENS

GEOSPATIAL BIG DATA ANALYTICS

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Open-Vocabulary Temporal Change Retrieval

Abstract

We build a semantic change search engine that, given a free-text query (e.g. “agricultural land converted to wetland”), ranks bi-temporal satellite image pairs by how well the change between the two timesteps matches the query, and returns the matched tiles, the timestep, a localisation heatmap, a match score, and a seasonal-vs-permanent flag. A query may name a change *type* – a property of the later state, such as “a new road” – or a directional change *transition* such as the example above, and the two are best served by different scoring approaches. The system uses frozen CLIP-variant vision-language backbones and compares three scoring approaches - a naive image baseline, a zero-shot Δ -similarity, and a parameter-efficient fine-tuned (PEFT) adapter - across three encoders on Dynamic EarthNet (DEN), with three further datasets (QFabric, LEVIR-CC/MCI, and SECOND-CC) validating the dataset-agnostic design and extending it to open-vocabulary breadth and pixel-level change localisation. All findings are reported under a strict statistical protocol (random-ranking baselines, FDR correction, and leave-one-AOI-out cross-validation). Out of distribution, the frozen zero-shot score beats the learned PEFT and LoRA adapters: the adapters’ high in-distribution scores (0.420–0.999 mAP) measure the adapter evaluated on its own training pairs (memorisation) and collapse to or below chance on held-out areas. Single-split test mAP on the 110-pair split is high-variance - the GeoRSCLIP+NRG configuration reaches 0.426 on one split, while its 5-fold AOI cross-validated estimate is 0.100 ± 0.139 . Defining retrieval relevance by the per-class pixel-fraction change the labels already encode, and scoring change locally with a patch-level Δ -score, raises cross-validated mAP to 0.193 ± 0.051 and makes localised change-types - new buildings and urban expansion, with new water borderline - retrievable above chance. On LEVIR-CC (a public benchmark of human-captioned building-change pairs) the same frozen engine retrieves salient building and road change strongly (per-query AP ≈ 0.6 –0.8) but subtle vegetation and sparse water change only near their prevalence floors, so the open-vocabulary signal scales sharply with change salience. SECOND-CC extends this to a seven-query open vocabulary -

every change type clears its prevalence floor but only modestly (zero-shot macro ≈ 0.33 vs a ≈ 0.30 floor). Using the LEVIR-MCI and SECOND-CC pixel masks we also score the change heatmap quantitatively: it is a weak localiser (pointing-game lift within ± 0.04 – 0.10 of a random-patch floor), so localising change is harder than retrieving it. The recurring lesson is that spectral physics and spatial locality carry what signal exists, while global embeddings and learned visual priors dilute or memorise it – and that single-split scores must be cross-validated before they estimate generalisation.

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1 Introduction and objectives

Traditional change detection is limited to a few predefined categories (e.g. “urban growth” or “forest loss”) [1]. This project moves toward open-vocabulary change retrieval by leveraging Vision-Language Models (VLMs) [2]: instead of training a class-specific detector, we build a system that identifies semantically defined transitions from natural-language queries (e.g. “new industrial buildings appearing on former agricultural land”) and searches across a multitemporal dataset to find the tiles and time-steps where the described change occurred. Text and image-change embeddings are aligned, and a pair is scored by the cosine similarity between its change embedding and the query embedding.

Per the GBDA Lab Project’s general guidelines, the work is constrained to a low-compute budget and must deliver a subset of an appropriate dataset, CLIP-variant frozen encoders, both zero-shot inference and light PEFT, retrieval metrics (Recall@ K , mAP), an error analysis of seasonal-vs-permanent confusion (“semantic drift”), and a Gradio [3] Semantic Change Search Engine as the product deliverable. The primary dataset is Dynamic EarthNet (DEN) [4], which provides daily multi-spectral Planet Fusion [5] data ideal for pinpointing the exact time-step of change (evaluated in Appendix B); the dataset-agnostic design is exercised on QFabric [6], LEVIR-CC/MCI [7, 8], and SECOND-CC [9].

This report makes three contributions. First, it delivers a working, dataset-agnostic open-vocabulary change-retrieval engine with a Gradio interface and a label-grounded Recall@ K /mAP benchmark. Second, it reports a careful, statistically audited comparison of frozen zero-shot scoring against light parameter-efficient fine-tuning across three CLIP-variant encoders, three colour modes, and four datasets, with leave-one-AOI-out cross-validation as the generalisation estimate. Third, it identifies and characterises a single organising principle that holds across every dataset: the recovered change signal scales with the spatial and semantic *salience* of the change. We state this salience hypothesis at the outset - salient, well-described construction change is retrievable, while subtle, sparse, or rare transitions collapse toward their prevalence floor - and test it systematically. The companion findings are that frozen RS-pretrained features with NIR false-colour and localised patch scoring carry what weak signal exists, and that low-compute fine-tuning does not improve retrieval out of distribution.

The remainder of this report covers the theoretical background (Section 2), related work (Section 3), the system architecture (Section 4), the retrieval approaches and scoring (Section 5), the data and encoders (Section 6), the evaluation protocol and metrics (Section 7), the experiments and results (Section 8), resources and operational metrics (Section 9), reproducibility (Section 10), limitations and future work (Section 11), and conclusions (Section 12).

2 Background and theory

Change detection: types and transitions. Remote-sensing change detection has classically estimated, per pixel or per region, whether and how land cover changed between two co-registered acquisitions [1]. Two questions are distinct and recur throughout this report. A change *type* is a property of the later state (“this is now a road”), recoverable from the after-image alone; a change *transition* is directional (“cleared land became a building”), and requires comparing both timesteps. Open-vocabulary retrieval generalises the classical setting by replacing a fixed label set with free-text queries, so the same engine must serve both questions; which scoring rule wins depends on which question a query asks.

Contrastive vision-language models. CLIP [2] and its open reproduction OpenCLIP [10] train a dual encoder - a vision tower (here a Vision Transformer [11]) and a text tower - so that matching image-text pairs have high cosine similarity - the dot product of the L_2 -normalised embeddings, $\cos(u, v) = u \cdot v / (\|u\| \|v\|) \in [-1, 1]$, with 1 meaning identical direction - in a shared embedding space, using a contrastive InfoNCE [12] objective over large web corpora. Because the space is shared and normalised, an unseen text query can be scored against an image by cosine similarity with no

task-specific training; this is the zero-shot transfer property the engine relies on. We exploit it twice: directly (the after-image score) and on a change feature built from the two timesteps’ embeddings.

Remote-sensing foundation models and the domain gap. General web-pretrained CLIP embeds everyday scene appearance, which need not align with overhead multi-spectral imagery or with subtle land-cover transitions; this is a distribution-shift problem, and zero-shot transfer is known to degrade under natural shift [13]. RS-specialised contrastive models narrow the gap by continuing contrastive pre-training on remote-sensing image–text data: GeoRSCLIP [14] on the large RS5M corpus, and RemoteCLIP [15] on curated RS caption data. A central empirical question of this report is whether such domain pre-training, or light task adaptation, materially closes the gap for temporal change retrieval.

Parameter-efficient fine-tuning. When a frozen backbone underperforms, the low-compute alternative to full fine-tuning is to train a small number of added parameters. Adapter modules [16] insert tiny trainable bottlenecks while freezing the backbone; LoRA [17] instead learns low-rank updates to existing weight matrices, adding well under 1% of parameters. We use both: a small projection-head adapter over the change feature, and LoRA on the visual encoder’s feed-forward projections. PEFT trades capacity for lower catastrophic-forgetting risk and compute; as the results show, limited capacity does not by itself prevent memorising a small training set.

Spectral background: NIR false-colour. Vegetation reflects strongly in the near-infrared and absorbs in the red, the basis of vegetation indices such as NDVI [18]. Mapping NIR-Red-Green into the three input channels (an “NRG” false-colour) therefore exposes vegetation and moisture contrasts that an RGB composite suppresses, which matters for the agriculture↔wetland transitions that dominate the primary dataset. The Planet Fusion [5] and Sentinel-2 [19] sensors both carry the requisite NIR band.

Retrieval metrics and the statistical protocol. Retrieval quality is measured with standard information-retrieval metrics [20]: for each query, average precision (AP) summarises the ranked list, mean average precision (mAP) macro-averages AP over queries, and Recall@ K is the fraction of relevant items in the top K . Two properties shape their interpretation here. A random ranker scores mAP equal to the query’s prevalence (its positive fraction), so mAP is read against a prevalence floor, not against zero; and Recall@ K is upper-bounded by K/R when the relevant set R is large, so it is read for ordering rather than absolute height. Because single small splits are high-variance, we estimate generalisation with k -fold *leave-one-AOI-out* cross-validation, which partitions by geographic area of interest so that no area appears in both train and evaluation, controlling the leakage that otherwise inflates learned adapters. Significance is assessed against a Monte-Carlo random-ranking baseline (a permutation test [21] on the ranking with the relevant set fixed), with bootstrap [22] confidence intervals and Benjamini–Hochberg [23] false-discovery-rate correction across the query set. Localisation of the change heatmap is scored with the pointing-game metric [24] - whether the most-activated patch falls on a true change region - and patch-level AP, each against a random-patch floor.

3 Related work

RS vision-language foundation models. GeoRSCLIP [14] and RemoteCLIP [15] adapt the CLIP recipe to remote sensing and are the domain backbones evaluated here; both target scene-level image–text alignment rather than temporal change, which is precisely the gap this project probes. Temporal earth-observation assistants such as TEOChat [25] pair imagery with instruction-style temporal questions; we reuse its QFabric question–answer annotations as a label source while keeping a lightweight retrieval engine rather than a generative assistant.

Change-captioning datasets. LEVIR-CC [7] pairs bi-temporal building-change scenes with human captions; LEVIR-MCI [8] adds pixel-level change masks over the same pairs; SECOND-CC [9] pairs captions with six-class SECOND [26] semantic maps for a far wider land-cover vocabulary. We use these not as captioning benchmarks but as open-vocabulary retrieval and localisation testbeds with caption- and mask-grounded relevance. The primary dataset, DynamicEarthNet [4], contributes dense daily multi-spectral imagery with monthly land-cover labels, which is what makes exact-timestep change retrieval possible; QFabric [6] and fMoW [27] contribute construction-centric and functional change taxonomies respectively.

Positioning. Against this body of work, the engine here is deliberately frozen-backbone and low-compute: it asks how far zero-shot scoring and light PEFT carry open-vocabulary change retrieval across four datasets under an honest statistical audit, rather than proposing a new trained change model. The reusable contribution is the dataset-agnostic protocol and the salience-scaled reading of when frozen VLM features succeed.

4 System architecture

For each bi-temporal pair (T_1, T_2) the pipeline proceeds as follows. A frozen VLM encodes both timesteps into L_2 -normalised global embeddings f_{T_1}, f_{T_2} , cached to disk per (dataset, encoder, split, colour mode). A change feature is then formed, either as a difference $\Delta f = f_{T_2} - f_{T_1}$ or as a concatenation $[f_{T_1}; f_{T_2}]$. The query text is encoded by the same model’s text tower into t , and the pair is scored by one of the approaches of Section 5. Figure 1 summarises the engine end to end.

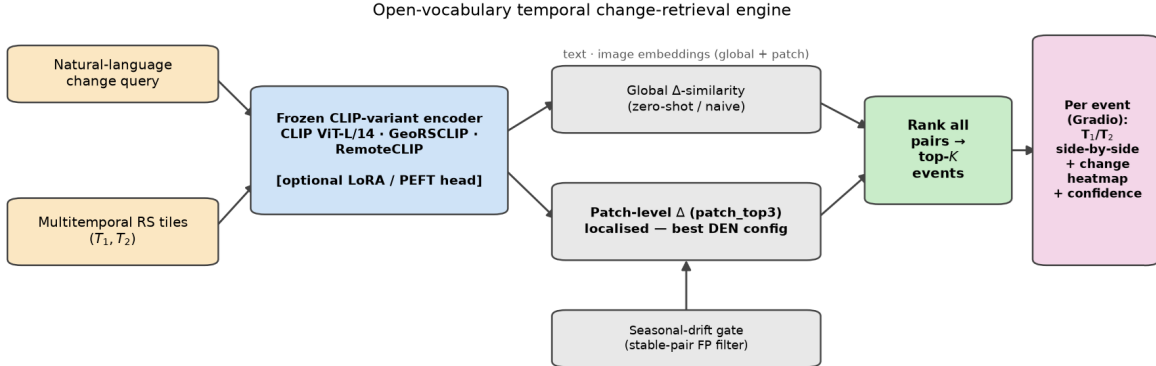


Figure 1: Open-vocabulary temporal change-retrieval engine. A frozen CLIP-variant encoder (CLIP ViT-L/14, GeoRSCLIP, or RemoteCLIP; optionally with a light LoRA/PEFT head) embeds the text query and both timesteps; a pair is scored by global Δ -similarity or by the localised `patch_top3` change score (the best Dynamic EarthNet configuration); pairs are ranked into the top- K change events, each surfaced in the Gradio app as a T_1/T_2 side-by-side with a query-conditioned change heatmap and a match score in $[0, 1]$. A seasonal-drift gate filters stable-pair false positives.

Dataset-agnostic core. A structural `TemporalDataset` protocol defines the only contract downstream code depends on: list the locations and bi-temporal pairs, load the images, load metadata, and derive a ground-truth change label. A registry maps a dataset name to a loader, and one concrete loader per dataset and on-disk format implements that contract – for Dynamic EarthNet (both the preprocessed 24-month arrays and the monthly rasters), for QFabric (polygon-centred crops carrying the real change-type labels benchmarked in Section 8.5, plus a status-transition sibling), and for the captioned, mask-annotated LEVIR-CC, LEVIR-MCI, and SECOND-CC. Adding a dataset or encoder

means adding a file, never editing the shared pipeline: the abstraction is enforced by the protocol, and the full source is in the linked repository.

Encoders. An `ImageTextEncoder` protocol has three implementations: `clip_vitl14` (HuggingFace CLIP [2] on a ViT [11] backbone, 768-d), `georsclip` (`open_clip` [10] ViT-B/32 + RS5M checkpoint [14], 512-d), and `remoteclip` (`open_clip` ViT-L/14 + RemoteCLIP checkpoint [15], 768-d). All backbones are frozen and run in evaluation mode; the embedding dimension drives cache and index sizing automatically.

PEFT training. Only the `ProjectionHead` adapter trains; the backbones stay frozen. Supervision is DEN’s weak caption derived from its land-use / land-cover (LULC) labels (e.g. “agriculture replaced by wetlands”, “stable forest land cover”). The loss is a masked symmetric InfoNCE [12]: pairs sharing an identical caption are treated as mutual positives [28], which avoids the false-negative problem of plain diagonal InfoNCE, since DEN captions repeat heavily.

Search engine. Evaluation is label-grounded against a fixed query set, and the same scoring path drives the Gradio Semantic Change Search Engine. A user types a free-text query (for example “new buildings or houses constructed”); the engine scores every bi-temporal pair in the loaded corpus, ranks them, and returns the top change events. Each event shows the T_1 (before) and T_2 (after) image side by side, a query-conditioned change heatmap highlighting where in the tile the queried change is strongest, a match score in $[0, 1]$, and a seasonal-vs-permanent note that flags pairs whose difference is driven by seasonal or illumination drift rather than persistent change. The match score is the raw retrieval score (Section 5) min-max normalised across the returned set: it ranks results relative to one another within a single query and is not a calibrated probability. The heatmap is the per-patch query-conditioned Δ -similarity field of Section 5, normalised the same way. Two optional re-ranking toggles (location diversity and geographic coherence) and a continental filter aid exploratory browsing. A snapshot is shown in Figure 2.

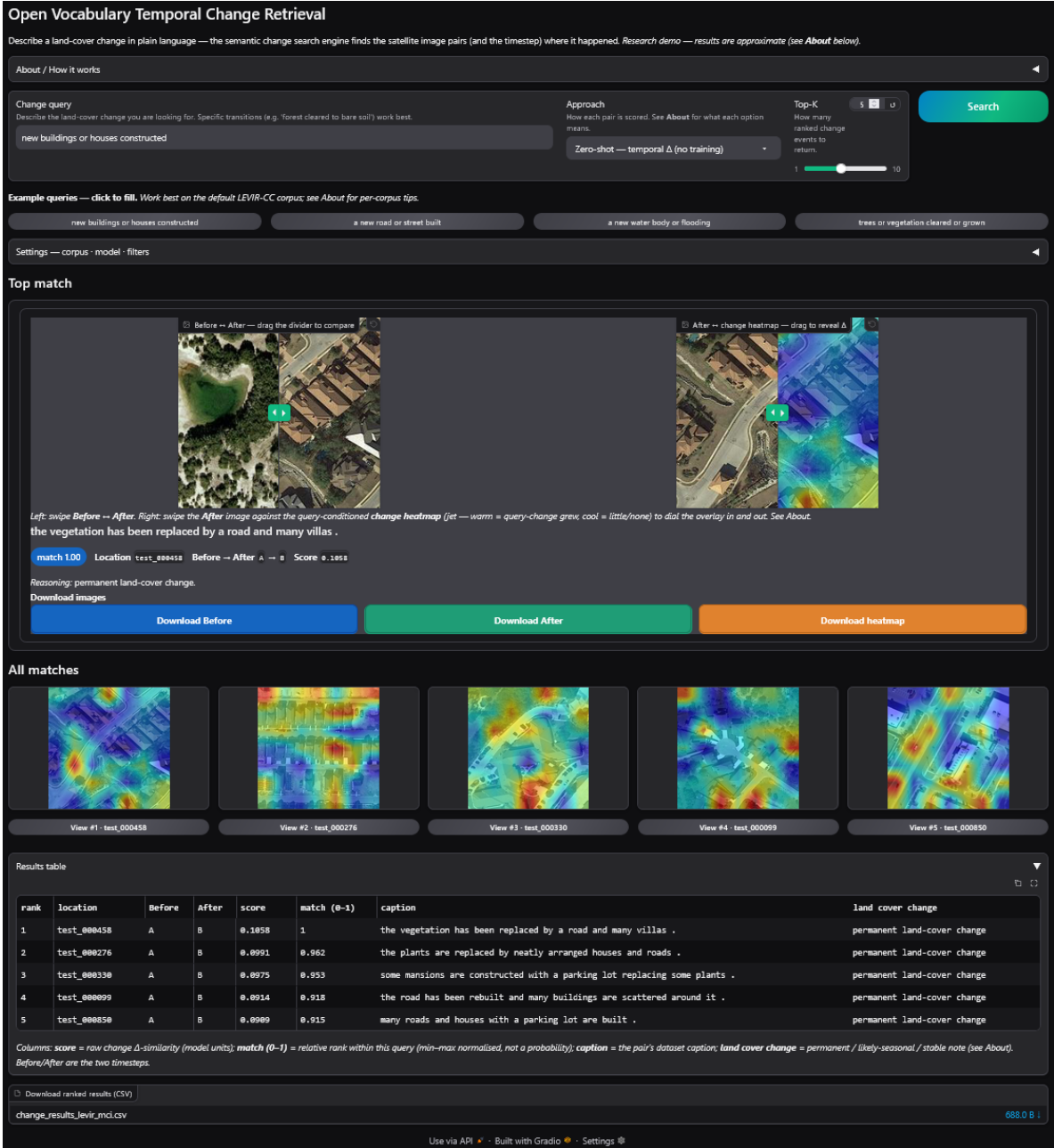


Figure 2: The deployed Gradio Semantic Change Search Engine. A free-text query returns a ranked list of change events; the top match is inspected through interactive before/after and after/heatmap comparison sliders, and the remaining matches through a thumbnail grid with per-result inspect controls, each result carrying a match score and a seasonal-vs-permanent note.

5 Retrieval approaches and scoring

Three global-embedding approaches are compared end-to-end, and a localised patch-level scorer is added on top. Let t be the L_2 -normalised query embedding and f_{T_1}, f_{T_2} the L_2 -normalised image embeddings of the two timesteps. The global scores are:

$$\begin{aligned}
s_{\text{naive}} &= \cos(t, f_{T_2}) && (\text{image-retrieval lower bound; no training}) && (1) \\
s_{\text{zero_shot}} &= \cos(t, f_{T_2}) - \cos(t, f_{T_1}) && (\Delta\text{-similarity; no training}) && (2) \\
s_{\text{peft}} &= \cos(t, g(\Delta f)) && (g = \text{ProjectionHead adapter, } \sim 0.5 \text{ M params}) && (3)
\end{aligned}$$

The naive baseline ignores the temporal dimension entirely (it scores the “after” image against the query) and therefore targets change *type*. The zero-shot score measures whether the query becomes more present from T_1 to T_2 and therefore targets change *transition*. The PEFT score passes a change feature through a small learned adapter g trained on weak captions. All three are unbounded model-unit values – a difference of cosine similarities lies in $[-2, 2]$ – and the retrieval metrics rank pairs directly by this raw score; the app additionally min-max normalises it per query into the $[0, 1]$ match score used for display only.

Change-feature mode. The adapter consumes a change feature built from the pair embeddings, in one of two modes: **difference** ($\Delta f = f_{T_2} - f_{T_1}$, dimension D) or **concatenate** ($[f_{T_1}; f_{T_2}]$, dimension $2D$, which keeps both endpoints). Unless otherwise stated all results use **difference**; the two modes are compared in Section 8.3. Figure 3 contrasts the three scoring modes, all of which share the single frozen encoder.

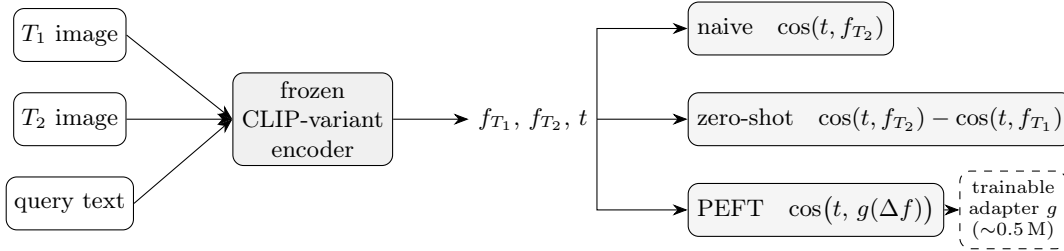


Figure 3: The three retrieval scores share one frozen CLIP-variant encoder. Only the PEFT branch trains a small adapter g over the change feature Δf (in **difference** or **concatenate** mode); the naive and zero-shot scores require no training. The naive score ignores T_1 entirely (change type), whereas the zero-shot Δ -similarity measures how much more present the query becomes from T_1 to T_2 (change transition).

Localised patch scoring. A global Δ over two whole-tile embeddings dilutes a change confined to a small region. The localised scorer therefore embeds each timestep into a grid of per-patch embeddings (dense features from the frozen encoder’s patch tokens, following the dense-CLIP recipe [29]) and computes a per-patch Δ -similarity $\cos(t, P_{2p}) - \cos(t, P_{1p})$; the pair score **patch_top3** is the mean of the top-3 per-patch deltas (**scripts/patch_eval.py**). The same per-patch Δ -field, conditioned on the query, is the change heatmap shown in the app and scored against masks in Section 8.5. Variants of the aggregation (soft-attention, spatial smoothing, a query-geometry gate) and a global/patch fusion are evaluated in Section 8.3.

LoRA adaptation. As an alternative to the projection-head adapter, LoRA [17] adapts the visual encoder’s feed-forward projections in place (Section 8.3), keeping the backbone otherwise frozen and adding under 0.5% of parameters.

6 Data and encoders

Dynamic EarthNet (primary). DEN is served from the ~ 7 GB DynNet preprocessed subset (in preference to the ~ 525 GB raw mirror): per-AOI daily RGB JPEGs (≈ 730 frames per AOI) plus

`labels/<AOI>.npy` of shape (24, 1024, 1024) monthly LULC, with a `split.json` (55 train / 10 val / 10 test AOIs). The 24 monthly label maps form the change timeline; each month maps to a representative daily RGB frame. These preprocessed label maps agree with the official torchgeo [30] Dynamic EarthNet raster masks (`torchgeo/dynamic_earthnet`) to 99.9999% per-pixel class agreement on the AOI tiles present in both releases (3 AOIs, 75.5 M valid pixels), confirming the preprocessing preserves the published ground truth; the preprocessed subset additionally retains the held-out test labels the official release withholds. A deterministic synthetic DEN fixture (`scripts/make_den_fixture.py`) reproduces the on-disk layout and an engineered label signal (urban growth, deforestation, seasonal snow-melt, stable negatives), so the full pipeline is testable in seconds with no network.

DEN also provides an on-disk near-infrared band (`_infra.jpeg`, one grayscale frame per daily timestep), exposed through three colour modes: `rgb` (standard 3-channel), `nrg` (NIR-Red-Green false-colour) and `ndvi` (single-band NDVI replicated across three channels). The working corpus is 605 bimonthly pairs on the train split, and 110 pairs each on val and test.

QFabric (secondary). QFabric [6] validates the dataset-agnostic design on a different taxonomy, sensor, and crop scale, using the polygon-centred crops from `jirvin16/TEOChatlas` [25] with the real QFabric change-type labels (the TEOChatlas RQA2 answers; 27,879 labelled crops) and, separately, the RQA5 per-timepoint development-status answers for a temporal transition task.

LEVIR-CC (human change captions). LEVIR-CC [7] provides a fundamentally different label source: 10,077 bi-temporal building-change pairs (256×256 , 0.5 m), each with five human-written change captions and a binary change flag (5,038 changed / 5,039 unchanged). The captions are parsed into open-vocabulary change tags (building, road, demolition, vegetation, water) so that free-text queries map directly to caption-grounded relevance, testing the engine on salient, well-described change in genuine human language and complementing DEN’s subtle spectral transitions and QFabric’s categorical types.

LEVIR-MCI (localisation masks on the LEVIR pairs). LEVIR-MCI [8] is a strict superset of LEVIR-CC - the same 10,077 pairs and captions, plus a pixel-level building/road change mask per pair. It is stored once and serves both `levir_cc` and `levir_mci`; the masks make the change-heatmap deliverable quantitatively measurable (Section 8.5).

SECOND-CC (open-vocabulary breadth + semantic masks). SECOND-CC [9] pairs 6,041 bi-temporal scenes with 30,205 human change captions and six-class SECOND [26] semantic maps for both phases (ground, tree, low-vegetation, water, building, playground). It is the breadth counterpart to LEVIR-CC, whose change is almost entirely building/road: SECOND-CC’s captions span a far wider land-cover vocabulary, and its per-phase semantic maps give both retrieval relevance (caption tags) and localisation ground truth (per-class and directed-transition change masks), through the same protocol.

Encoders. Table 1 lists the three frozen CLIP-variant encoders. All embeddings are L_2 -normalised; the dimension automatically drives cache and index sizing.

Table 1: The three frozen vision-language encoders.

Encoder	Backbone	Pre-training	Dim
<code>clip_vitl14</code>	ViT-L/14 (HuggingFace CLIP)	general (CLIP)	768
<code>georsclip</code>	ViT-B/32 (<code>open_clip</code>)	RS5M (remote sensing)	512
<code>remoteclip</code>	ViT-L/14 (<code>open_clip</code>)	RemoteCLIP	768

7 Evaluation protocol and metrics

Evaluation is label-grounded: a fixed query set is defined per dataset, each query maps to a relevance rule over the derived `PairLabel`, and the ranked corpus is scored with the information-retrieval metrics of Section 2.

Relevance. For the captioned and masked datasets (LEVIR-CC, SECOND-CC, QFabric), relevance is the caption or change-type predicate that a query names. For DEN, relevance is derived from the per-class *pixel-fraction* change the monthly LULC labels already encode: a pair is a positive for a change-type query when at least 5% of its pixels undergo that class transition. This pixel-fraction rule makes 9 of 10 DEN change-types evaluable (new buildings 14 positives, deforestation 13, soil 25, water 9; only snow is genuinely absent from the subset). A coarser alternative - counting a pair only when the dominant class of the whole 1024^2 tile flips - is satisfied by just 8.6% of pairs (71/825), 44 of them wetland $\leftrightarrow$ agriculture, and is too sparse to evaluate most queries; pixel-fraction relevance is therefore the reference rule for DEN, and the two are contrasted directly in Section 8.1.

Metrics. We report per-query and macro Recall@ K , mAP, and a seasonal-drift figure (the fraction of non-relevant top- K retrievals that involve snow/ice - seasonal events wrongly returned for permanent-change queries). mAP is macro-averaged only over queries with at least one positive in the corpus, so the evaluable query set can differ per split. Two interpretive caveats follow from Section 2: a random ranker scores $\text{mAP} \approx \text{prevalence}$ (0.17 on QFabric-TEO, 0.08 on DEN-test), so mAP is read against that floor; and Recall@ K on large relevant sets is ceiling-bounded (with 480 relevant pairs, $R@10 \leq 10/480 = 0.021$), so QFabric is judged by mAP only. Change-heatmap localisation is scored on the LEVIR-MCI and SECOND-CC masks by the pointing-game [24] metric and patch-AP, each against the encoder’s own random-patch floor.

Cross-validation and significance. Because a single 110-pair split is high-variance, the generalisation estimate is 5-fold *leave-one-AOI-out* cross-validation over the merged 75-AOI, 825-pair corpus, partitioning by AOI so no area appears in both train and evaluation. For learned adapters this leakage control is essential: an adapter evaluated on areas it trained on measures memorisation, not retrieval. Significance is assessed against a Monte-Carlo random-ranking baseline (the relevant set fixed, the ranking shuffled) [21], with bootstrap [22] confidence intervals and Benjamini-Hochberg [23] FDR correction across the query set; a query is reported as significant only after FDR correction.

8 Experiments and results

All headline experiments use real Dynamic EarthNet. The fast test suite (mock encoders, synthetic fixture, no network) passes; the full suite additionally runs the real-CLIP text-encoder test (skipped unless its weights are present) and covers the embeddings cache and round-trip, retrieval (naive/zero_shot/peft), benchmark metrics, PEFT training (loss decreases, save/load), the encoder protocol/registry, and the app’s query path (real fixture tiles + heatmap + seasonal note). We report the cross-validated DEN result first (Section 8.1), since it is the generalisation verdict, then the evidence that learned adapters memorise (Section 8.2), the ablations that locate the signal (Section 8.3), robustness (Section 8.4), and the cross-dataset validation that establishes the salience law (Section 8.5).

8.1 Main result on Dynamic EarthNet

Under the cross-validated protocol of Section 7, the defensible DEN result is narrow but honest: a frozen, RS-pre-trained encoder (GeoRSCLIP) with NIR false-colour and localised patch scoring retrieves construction change (new buildings, urban expansion) and large-area wetland change above chance,

with new water borderline, at ≈ 0.20 cross-validated mAP; no learned adapter beats it on held-out areas. Table 2 traces the GeoRSCLIP+NRG estimate across the relevance rule and the scoring choice.

Table 2: GeoRSCLIP NRG zero-shot on DEN: single-split mAP is high-variance, while the cross-validated estimate is the generalisation verdict. “S1” = pixel-fraction relevance; “S3” = patch-level scoring. Random baseline ≈ 0.08 .

Evaluation	test split	full corpus	5-fold CV
dominant-flip relevance	0.426	0.037	0.100 ± 0.139
+ fraction relevance (S1)	-	0.091	0.139 ± 0.024
+ patch top-3 scoring (S3)	-	0.122	0.193 ± 0.051

Single-split variance. The widely-quoted 0.426 mAP for GeoRSCLIP+NRG is a single 110-pair test split on which the cross-validation folds span 0.032–0.348; that split coincides with the easy high-wetland fold. The 5-fold AOI cross-validated estimate of the same configuration is 0.100 ± 0.139 (full-corpus 0.037), so the single split overstates generalisation by roughly twofold and only the cross-validated figure should be read as the result.

Relevance and locality. The pixel-fraction relevance rule (Section 7) roughly triples full-corpus mAP and tightens the cross-validation interval to 0.139 ± 0.024 by making nine change-types evaluable rather than three. Beyond relevance, the remaining ceiling is the global embedding: differencing two whole-tile 1024^2 embeddings dilutes a small change region, leaving localised change-types (new buildings, urban expansion, water) near chance. Scoring from per-patch embeddings with **patch_top3** (Section 5) lifts cross-validated mAP to 0.193 ± 0.051 - the best DEN configuration - and makes new buildings ($q=0.003$) and urban expansion ($q=0.006$) beat random under FDR correction for the first time — four of the nine change-types are FDR-significant in this configuration — with new water borderline ($q=0.053$) (Figure 4). Diffuse large-area wetland change still prefers the global score, so the two are complementary. GeoRSCLIP’s coarse 49-patch grid beats CLIP-L/14’s finer 256-patch grid (0.193 vs 0.149), so domain pre-training outweighs patch resolution.

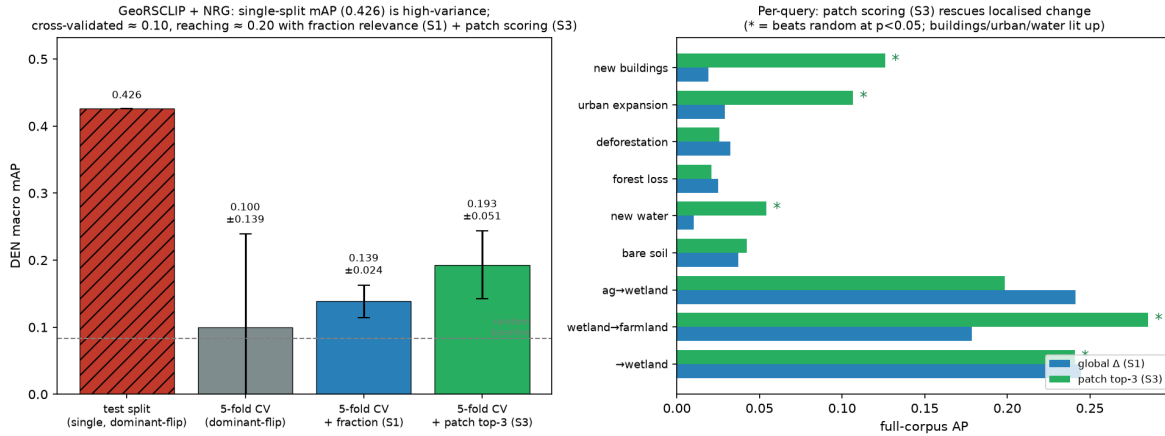


Figure 4: DEN result decomposition (GeoRSCLIP+NRG). Left: the high-variance single-split mAP versus the cross-validated estimate, and the effect of pixel-fraction relevance (S1) and localised patch scoring (S3), which together reach ≈ 0.20 cross-validated mAP. Right: per-query, patch-level scoring (S3) recovers localised change-types (buildings/urban/water) that global pooling leaves at chance (* = beats random, $p < 0.05$).

The frozen adapter does not help out of distribution. Leakage-free cross-validated PEFT on GeoRSCLIP+NRG (0.196 ± 0.049 , fraction relevance) overlaps frozen zero-shot (0.139 ± 0.024) within fold variance - neither a collapse nor a gain - so low-compute fine-tuning is not shown to improve retrieval out of distribution. The high in-distribution PEFT figures (0.42–0.998) are the adapter scored on its own training pairs, as Section 8.2 establishes directly.

8.2 Learned adapters memorise the training areas

The in-distribution and cross-split measurements isolate why the adapters do not generalise. Table 3 reports the train split, where the PEFT column is the adapter evaluated on the very pairs it was trained on, so it measures memorisation capacity rather than retrieval.

Table 3: Training-set fit (train split, 605 pairs, RGB, **difference**) - PEFT scored on its own training pairs (memorisation), not a held-out result.

Encoder	naive	zero-shot	PEFT	PEFT R@10
CLIP ViT-L/14 (768-d)	0.031	0.043	0.420	0.36
GeoRSCLIP ViT-B/32 (512-d)	0.027	0.040	0.335	0.26
RemoteCLIP ViT-L/14 (768-d)	0.024	0.057	0.352	0.30

Three observations follow. First, zero-shot is near chance in-distribution: Δ -similarity beats the naive image baseline only marginally, because CLIP and GeoRSCLIP embed scene appearance rather than directional land-cover transition, so differencing two normalised global embeddings discards the localised change signal. Second, the $\sim 8\text{--}10\times$ PEFT lift shows the adapter memorises the 55 training AOIs (the same train fit reaches mAP 0.998 on QFabric, Section 8.5), not that it retrieves. Third, this train-fit ranking reverses on held-out data, where GeoRSCLIP leads (Table 4).

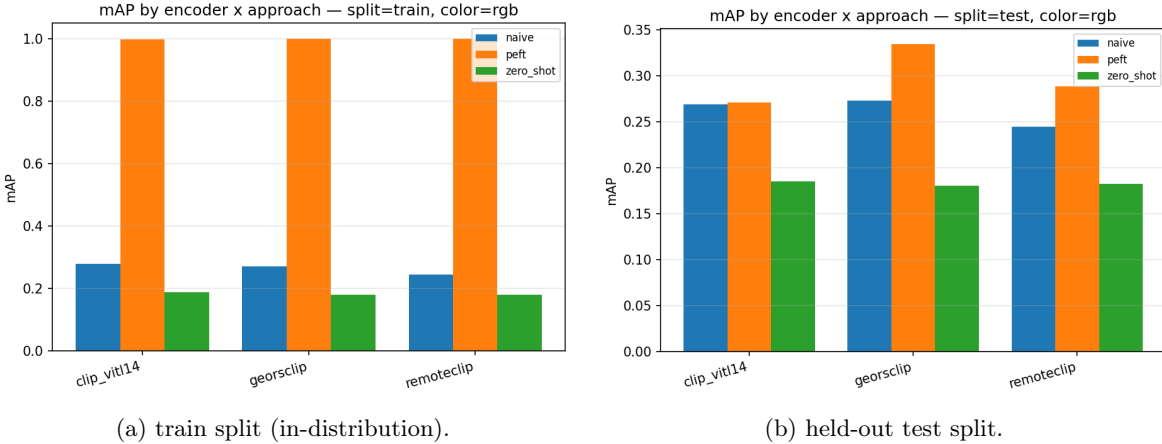


Figure 5: mAP by encoder $\times$ approach (RGB). The large PEFT bars on train (a) collapse to the frozen scores on the held-out test split (b), where the training-free baselines (GeoRSCLIP zero-shot in particular) are competitive - the memorisation signature.

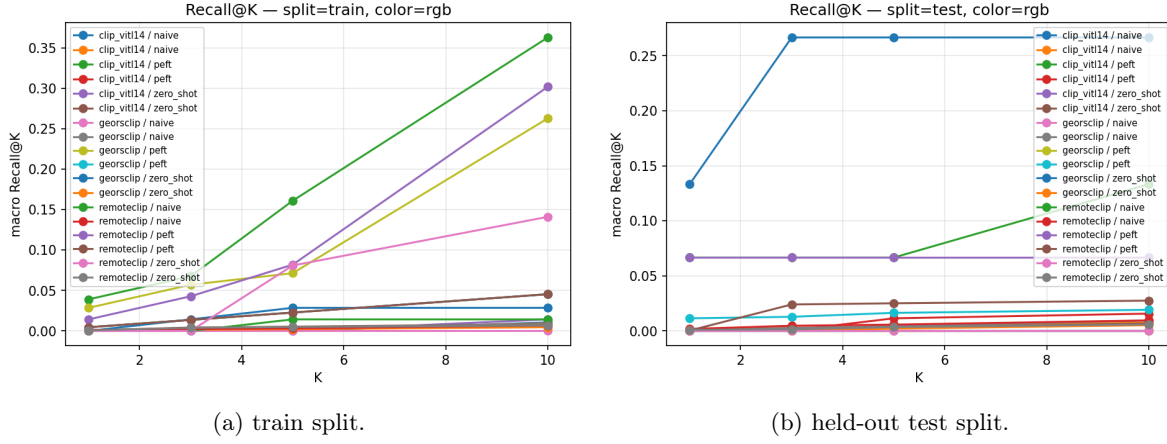
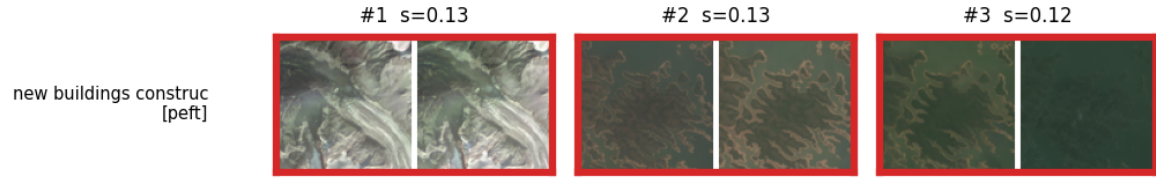


Figure 6: Macro Recall@ K versus K for every encoder $\times$ approach (RGB), the brief’s second retrieval metric alongside mAP. Recall@ K is bounded by the large relevant-set sizes, so the curves are read for relative ordering, not absolute height.

Zero-shot vs PEFT — top-3 (clip_vitl14, split=train)
[T1 | T2]; green=relevant, red=not



Zero-shot vs PEFT — top-3 (clip_vitl14, split=train)
[T1 | T2]; green=relevant, red=not

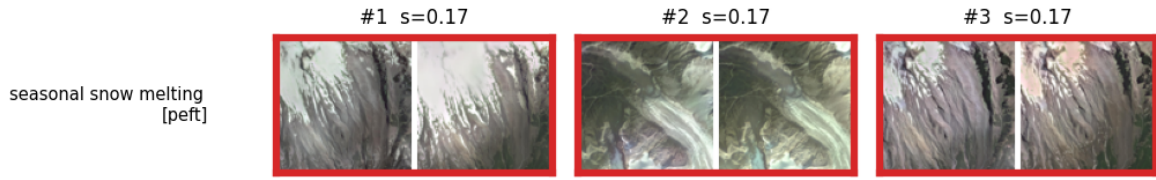
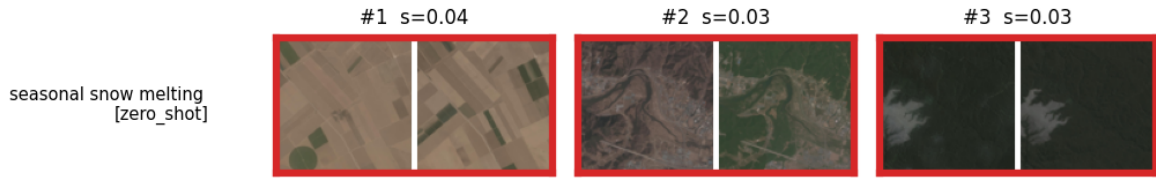


Figure 7: Qualitative zero-shot vs PEFT (CLIP, train): top-3 retrievals per query, $[T_1 | T_2]$, green = relevant. The adapter sharpens scores, but the gain is concentrated on the classes its weak captions cover.

Training the adapter on the train split and evaluating across train/val/test makes the collapse explicit (Table 4). On unseen val and test AOIs the adapter is equal to or worse than zero-shot, while GeoRSCLIP zero-shot reaches 0.299 mAP on the test split with no training at all.

Table 4: Cross-split mAP (adapter trained on train; RGB, **difference**).

Encoder	approach	train	val	test
CLIP ViT-L/14	naive	0.031	0.053	0.046
CLIP ViT-L/14	zero-shot	0.043	0.051	0.043
CLIP ViT-L/14	PEFT	0.420	0.042	0.040
GeoRSCLIP ViT-B/32	naive	0.027	0.030	0.061
GeoRSCLIP ViT-B/32	zero-shot	0.040	0.036	0.299
GeoRSCLIP ViT-B/32	PEFT	0.335	0.087	0.041
RemoteCLIP ViT-L/14	naive	0.024	0.029	0.121
RemoteCLIP ViT-L/14	zero-shot	0.057	0.025	0.050
RemoteCLIP ViT-L/14	PEFT	0.352	0.028	0.103

8.3 Ablations: what carries the signal

Colour mode. Three colour modes are compared (zero-shot mAP, all three splits) in Table 5. NRG is the best colour mode on held-out test for every encoder.

Table 5: Colour-mode ablation (zero-shot mAP). NRG is the best colour mode on held-out test for every encoder. Best test per encoder in bold.

Encoder	color	train	val	test
CLIP ViT-L/14	RGB	0.043	0.051	0.043
CLIP ViT-L/14	NRG	0.033	0.066	0.104
CLIP ViT-L/14	NDVI	0.032	0.045	0.064
GeoRSCLIP	RGB	0.040	0.036	0.299
GeoRSCLIP	NRG	0.025	0.030	0.426
GeoRSCLIP	NDVI	0.028	0.065	0.216
RemoteCLIP	RGB	0.057	0.025	0.050
RemoteCLIP	NRG	0.023	0.047	0.129
RemoteCLIP	NDVI	0.022	0.048	0.055

NRG hurts on train (-0.010 to -0.034) but substantially helps on test ($+0.061$ CLIP, $+0.127$ GeoRSCLIP, $+0.079$ RemoteCLIP). NDVI usually lands between NRG and RGB (though for GeoRSCLIP its single-channel collapse drops it below even RGB on test, 0.216 vs 0.299): NDVI provides some NIR signal but collapses all spectral texture into a single channel replicated across R/G/B, so the RS-pretrained encoders cannot exploit the inter-channel colour contrasts that NRG preserves. RemoteCLIP NRG (test 0.129) improves over RGB (0.050) but lags GeoRSCLIP NRG, whose RS5M pre-training gives a stronger prior for the NIR-Green spectral contrast characteristic of vegetation transitions. The single-split NRG lift for GeoRSCLIP (0.426) is a high-variance fold (cross-validated 0.100 ± 0.139 , Section 8.1); NRG remaining the best colour mode for every encoder, and the encoder ordering (GeoRSCLIP $\gg$ CLIP $\approx$ RemoteCLIP), both hold under cross-validation.

Change-feature mode: difference vs concatenate. Re-training the adapter under **concatenate** (same recipe, RGB, 40 epochs) and comparing PEFT mAP across splits (Table 6) shows that keeping both endpoints overfits less.

Table 6: PEFT change-feature mode: difference vs concatenate (RGB). Best per encoder/column in bold.

Encoder	mode	train	val	test
CLIP ViT-L/14	difference	0.420	0.042	0.040
CLIP ViT-L/14	concatenate	0.370	0.081	0.050
GeoRSCLIP	difference	0.335	0.087	0.041
GeoRSCLIP	concatenate	0.275	0.086	0.070
RemoteCLIP	difference	0.352	0.028	0.103
RemoteCLIP	concatenate	0.359	0.109	0.078

Concatenate consistently lowers in-distribution train mAP (CLIP -0.050 , GeoRSCLIP -0.060) but raises held-out val for all three encoders and test for two of three: keeping both endpoints rather than collapsing to their difference preserves absolute land-cover context the adapter can use to generalise, at the cost of training-set fit. The effect is modest and PEFT still does not beat frozen zero-shot out of distribution, but **concatenate** is the better-generalising change feature.

LoRA on the visual encoder. LoRA [17] applied to GeoRSCLIP’s ViT-B/32 visual encoder targets the FFN projections `c_fc` and `c_proj` in each ResBlock: 368,640 trainable params out of 88M (0.42%). The attention output projection `out_proj` is deliberately not adapted - `open_clip`’s attention is an `nn.MultiheadAttention` whose forward reads `out_proj.weight` directly (via `F.multi_head_attention_forward`) and never calls `out_proj.forward`, so a LoRA wrapper there receives no gradient. Training is online (images loaded on the fly) using the same masked symmetric InfoNCE loss as the ProjectionHead. Table 7 compares frozen zero-shot against LoRA-adapted zero-shot (GeoRSCLIP + NRG, rank=4, $\alpha=8$, 20 epochs).

Table 7: LoRA (GeoRSCLIP + NRG) vs frozen zero-shot. LoRA fits train and matches val but collapses on the held-out test split. Best per row in bold.

split	zero_shot (frozen)	LoRA zero_shot
train	0.025	0.153
val	0.030	0.034
test	0.426	0.071

LoRA fits the train split ($0.025 \rightarrow 0.153$) but collapses out of distribution (0.071 on test). That test 0.071 is numerically just above ProjectionHead PEFT’s RGB test (0.041), but the comparison is cross-colour-mode and both sit at or below the DEN-test random floor (≈ 0.083) - neither learned adapter beats chance on held-out AOIs. A rank/epoch sweep (Table 8; `scripts/lora_sweep.py`, in-memory) finds no capacity sweet spot.

Table 8: LoRA rank/epoch sweep (GeoRSCLIP + NRG, zero_shot mAP). Every configuration fits train yet collapses on test - no capacity sweet spot.

rank	α	epochs	train	test
4	8	20	0.153	0.071
8	16	20	0.168	0.070
16	32	20	0.135	0.058
8	16	40	0.146	0.059

Every rank/epoch setting fits the train split (0.13–0.17, well above the frozen 0.025) yet overfits to test 0.06–0.07; more rank or more epochs only memorise harder. This reinforces the project’s core finding that spectral physics (NRG false-colour) generalises while learned visual priors only memorise.

Aggregation refinements (honest negatives). Several cheaper in-scope refinements were tested and none improved on `patch_top3`, reported here as honest negatives. Averaging the query embedding over a prompt ensemble is a wash (0.142 cross-validated mAP); fusing the global Δ -cosine with the patch score by z-scoring and summing hurts (0.165), the mostly-noise global signal diluting the stronger patch signal; and two training-free change-attention variants - query-conditioned soft-attention over the per-patch Δ (0.137) and 3×3 spatial smoothing of the Δ -map (0.168) - refine the aggregation without breaking the ≈ 0.20 ceiling. Routing each query by its a-priori spatial geometry - diffuse phrasings to the global score, localised ones to patch top-3 - scores 0.186 ± 0.051 under the same cross-validation, statistically indistinguishable from ungated patch top-3 and recovering the same four FDR-significant queries, so query geometry is not a usable routing signal at this encoder scale. A learned query-to-patch attention head is deliberately not pursued, since every trained head on this data memorises the training AOIs with no held-out gain (Section 8.2).

8.4 Robustness: seasonal vs permanent

The benchmark reports seasonal drift@ K (non-relevant top- K retrievals that involve snow/ice, for permanent-change queries). On the DEN train split this is 0.00 at all K for every encoder/approach - there is essentially no seasonal (snow/ice) class in this subset, so seasonal→permanent confusion does not arise here. The mechanism is nonetheless implemented and exercised on the synthetic fixture, which contains a seasonal snow-melt pair: the app flags it (“involves snow/ice - likely SEASONAL, not permanent”) and the benchmark counts it as drift if mis-retrieved.

The dominant real error is not seasonal confusion but low recall from class imbalance and weak label-derived captions (many near-stable bimonthly pairs; agri↔wetland visually subtle). The per-query confusion analysis makes this concrete: on the best test configuration (GeoRSCLIP + NRG), the wetland queries’ top-10 are dominated by **stable** pairs (6–7 of 10), so the failure mode is surfacing no-change pairs (Figure 8).

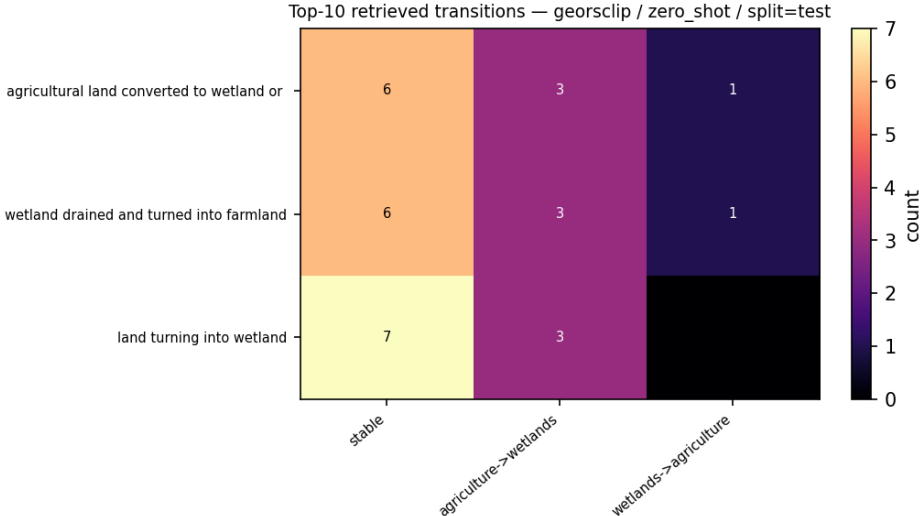


Figure 8: Per-query confusion (GeoRSCLIP + NRG zero-shot, test): top- K retrievals binned by actual label transition; wetland queries are dominated by **stable** pairs.

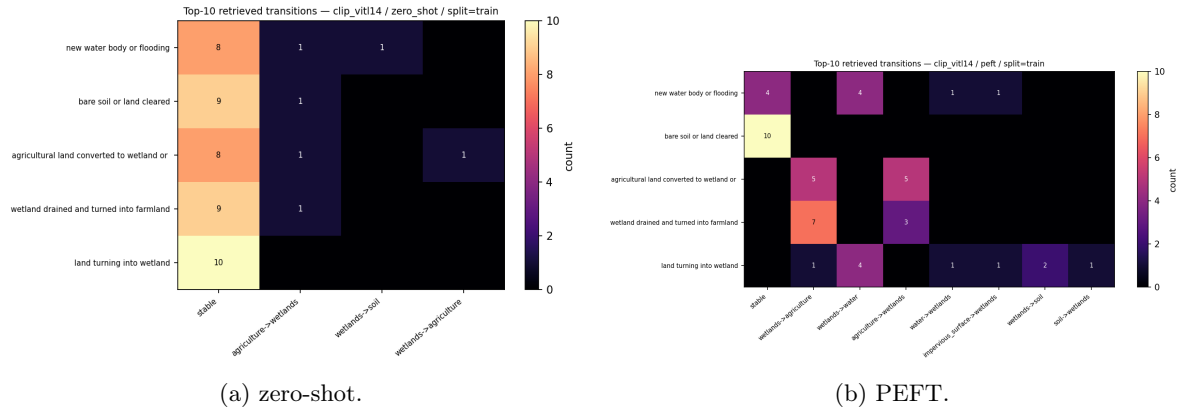


Figure 9: In-distribution confusion (CLIP ViT-L/14, train split): top- K retrievals binned by actual transition. The PEFT adapter (b) concentrates each query’s retrievals on its own positive transition far more sharply than zero-shot (a) - the visible face of the train-set memorisation of Section 8.2.

Seasonal robustness is also probed directly. The zero-shot score is turned into a whole-image binary gate: for a change query t and a pair’s L_2 -normalised global embeddings, the gate fires when $\Delta = \cos(t, f_{T_2}) - \cos(t, f_{T_1}) > \tau$. On a stable pair $f_{T_1} \approx f_{T_2}$, so $\Delta \approx 0$ for any query and every firing is a false positive; sweeping τ over the stable subset (DEN’s own `PairLabel.stable` flag) yields a false-positive-rate curve. Table 9 reports the curve for the strongest configuration (GeoRSCLIP, RGB, test split, 24 stable pairs).

Table 9: Stable-pair false-positive rate of the Δ -similarity gate (GeoRSCLIP, RGB, test split; 24 stable pairs, mean stable $\Delta = 0.0125$).

mean stable Δ	FPR@0.00	FPR@0.02	FPR@0.05	FPR@0.10
0.0125	0.875	0.333	0.000	0.000

Stable pairs carry a near-zero mean Δ -similarity (0.0125), so the false-positive rate falls to zero once the threshold clears the noise floor, and is already zero at $\tau = 0.05$. The high 0.875 rate at $\tau = 0$ merely reflects that any nonzero Δ trips a zero threshold and is not a failure mode. The gate confirms that seasonal and illumination drift on stable pairs does not masquerade as change once a modest threshold is applied.

8.5 Cross-dataset validation and the salience law

The three further datasets test the dataset-agnostic design and the salience hypothesis of Section 1. QFabric establishes that task geometry sets the naive-versus- Δ trade-off; LEVIR-CC and SECOND-CC establish that recovery scales with change salience; the masks quantify that localisation is harder than retrieval.

QFabric change-type retrieval. Images are the polygon-centred crops from jirvin16/TEOChatlas; labels are the real QFabric change types (the RQA2 answers), joined to crops by the shared filename scheme. On a stratified ≤ 120 crops/class subset ($N = 2476$ pairs) with six change-type queries and frozen encoders (RGB), Table 10 reports mAP.

Table 10: QFabric change-type retrieval (frozen encoders, RGB, mAP). naive beats zero-shot - the opposite of DEN. Best per row in bold.

Encoder	naive	zero-shot
CLIP ViT-L/14	0.274	0.187
GeoRSCLIP	0.269	0.182
RemoteCLIP	0.233	0.180

Naive beats zero-shot here, the opposite of DEN. QFabric change types (residential / road / industrial ...) are identifiable from the after-image content alone, so $\cos(t, f_{T_2})$ wins; the directional Δ -similarity that helped on DEN here adds noise. The random-ranking baseline is the macro class prevalence, 0.167; **naive** (0.274) sits +0.11 above chance while **zero_shot** (0.187) is only +0.02. Per-query (CLIP naive), the signal concentrates in road (AP 0.50), residential (0.36) and industrial (0.30); demolition (0.25) and commercial (0.22) are weak; mega_projects (0.03) is at chance. The pipeline runs end-to-end on a different taxonomy, sensor, and crop scale with no code changes beyond a loader and a query set, confirming the dataset-agnostic abstraction and surfacing a regime in which after-image content outweighs the temporal Δ .

QFabric PEFT. Closing the zero-shot-vs-PEFT comparison on QFabric uses a stratified, crop-disjoint train/test split (train ≤ 120 crops/class, $N=2422$ pairs; held-out test $N=2048$), a ProjectionHead adapter trained on the train split with captions phrased like the queries (**difference**, 40 epochs), evaluated on both splits (Table 11).

Table 11: QFabric PEFT (mAP). Adapter overfits train (≈ 0.999) but on test sits at-or-above naive. Best per row in bold.

Encoder	split	naive	zero-shot	PEFT
CLIP ViT-L/14	train	0.278	0.189	0.998
CLIP ViT-L/14	test	0.269	0.185	0.271
GeoRSCLIP	train	0.271	0.180	0.999
GeoRSCLIP	test	0.272	0.181	0.334
RemoteCLIP	train	0.244	0.179	0.999
RemoteCLIP	test	0.245	0.183	0.288

PEFT overfits train on both datasets (≈ 0.999 here) but generalises differently. On held-out QFabric test the adapter is at-or-above **naive**: GeoRSCLIP +0.062 (0.334, the best QFabric result), RemoteCLIP +0.043 (0.288), CLIP neutral. This contrasts with DEN (Section 8.2), where PEFT test collapsed below zero-shot (GeoRSCLIP RGB: 0.299 $\rightarrow$ 0.041). The difference is what the adapter must learn: QFabric change types are consistent visual categories (a road looks like a road across crops), so the head learns transferable features; DEN’s directional spectral transitions are subtle and AOI-specific, so the head memorises training-AOI statistics. PEFT’s value is therefore dataset-dependent: harmful on DEN, mildly helpful on QFabric. Both effects are small in absolute terms, since a random ranker already scores $\text{mAP} \approx 0.17$ on QFabric.

QFabric status-transition retrieval: the regime inverts. QFabric also carries a temporal task through the TEOChatlas RQA5/RTQA5 answers: the per-timepoint development status of each region, on a nine-stage scale from greenland through land-cleared and construction-started to operational. Joining each answer to its frame (`scripts/build_qfabric_status_labels.py`, dataset `qfabric_status`) gives every pair a status transition, and six transition queries (for example “new building construction has started”) are relevant only to pairs whose status actually changes; stable pairs

become hard negatives with the same end-state. On a stratified subset of ≤ 120 crops per final status ($N = 4200$ pairs, frozen encoders, RGB), Table 12 reports mAP.

Table 12: QFabric status-transition retrieval (RQA5; frozen, RGB, mAP). The directional zero-shot score now matches or beats naive on every encoder, inverting the change-type regime of Table 10. Best per row in bold.

Encoder	naive	zero-shot
CLIP ViT-L/14	0.057	0.057
GeoRSCLIP	0.079	0.084
RemoteCLIP	0.064	0.072

The regime inverts: zero-shot is at least as good as naive on all three encoders (+0.000 CLIP, +0.005 GeoRSCLIP, +0.007 RemoteCLIP), the opposite of the change-type result, where naive led by about 0.09. The directional Δ is the appropriate tool for transitions, while after-image content is the appropriate tool for types - one dataset, two tasks, two regimes. The edge is consistent but small, and both scores sit just above the 0.043 macro-prevalence baseline: status-transition retrieval is genuinely hard zero-shot, with long irregular temporal baselines and many subtle or rare transitions. The signal concentrates in the visually distinct, well-populated transitions - land cleared (AP 0.17) and construction started (0.18) - and collapses on the rare or long-range ones (demolition 0.01, $n = 32$; vacant-to-finished building 0.02, $n = 64$). Relabelling the same crops along a temporal axis rather than a categorical one inverts which scoring approach wins, direct evidence that the naive-versus- Δ trade-off is set by task geometry rather than by the dataset.

LEVIR-CC: salience-scaled retrieval. LEVIR-CC pairs are parsed into five open-vocabulary change tags from their captions - building, road, demolition, vegetation, water - and one free-text query maps to each. On the held-out test split (1929 pairs, frozen encoders, RGB), Table 13 reports per-query average precision and the five-query macro. The macro random-ranking baseline is the mean query prevalence, 0.255 (building 45.2%, road 42.7%, demolition 14.8%, vegetation 23.9%, water 1.0%).

Table 13: LEVIR-CC open-vocabulary change retrieval (test split, 1929 pairs, frozen, RGB; zero-shot average precision per query, with the five-query macro mAP). Test positives per query in parentheses. Best per row in bold.

Query (positives)	GeoRSCLIP	CLIP ViT-L/14	RemoteCLIP
new buildings or houses (871)	0.804	0.725	0.830
a new road or street (824)	0.571	0.631	0.654
buildings demolished (286)	0.297	0.138	0.234
trees or vegetation (462)	0.159	0.182	0.186
a lake, pond, or water (19)	0.151	0.221	0.096
macro mAP (5 queries)	0.396	0.379	0.400

The five-query macro (0.38–0.40 zero-shot; naive 0.41–0.43) is held up by two strong queries and dragged by three weak ones, so the per-query reading is the honest one. Building and road change - large, high-contrast, and described in the vocabulary the CLIP text tower was trained on - are retrieved strongly (buildings ≈ 0.8 , roads ≈ 0.6), whereas demolition, vegetation, and the 19-positive water class sit at or only just above their prevalence floors. This is the salience-proportional pattern the project finds in every dataset, pooled and quantified in Figure 12. The RS-pre-trained encoders still gain from the directional Δ on the strong queries, the same type-versus-transition split seen on QFabric.

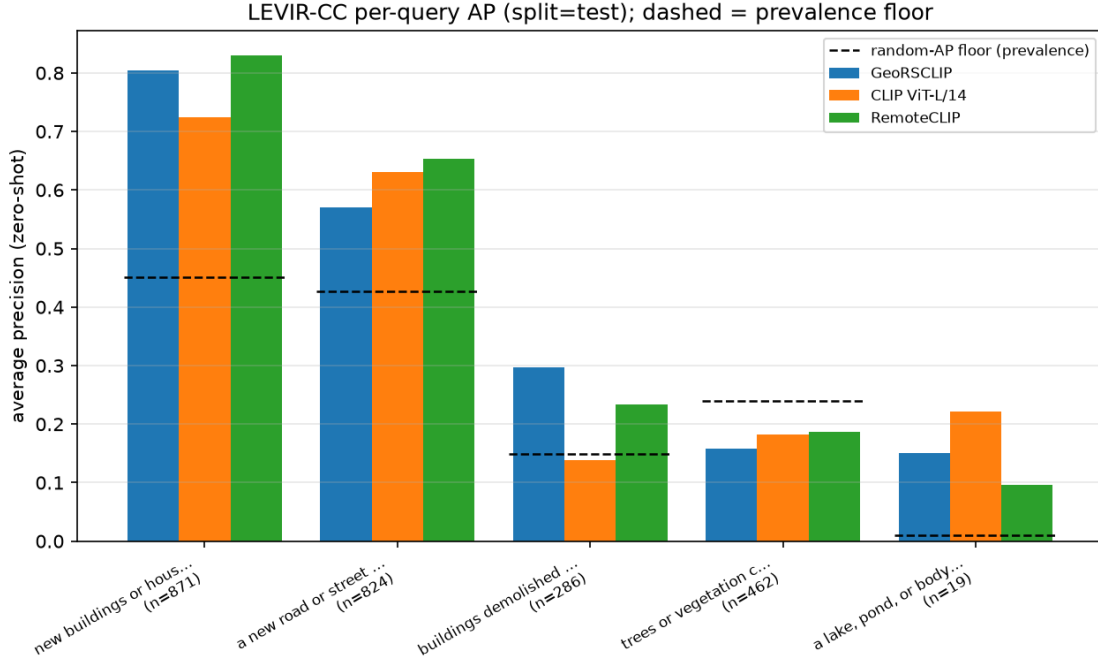


Figure 10: LEVIR-CC per-query zero-shot AP by encoder (test split, RGB), with each query’s prevalence floor (dashed). Salient building and road change clears the floor by a wide margin; demolition, vegetation, and the 19-positive water class sit at or only just above it.

SECOND-CC: open-vocabulary breadth. SECOND-CC [9] is the breadth counterpart: 6,041 pairs with human captions and six-class semantic maps for both phases. Seven free-text queries cover the six SECOND classes plus road. Test split, 1,227 pairs, frozen encoders, RGB; macro random-ranking floor 0.297.

Table 14: SECOND-CC zero-shot per-query AP (test split). Buildings dominate; every type clears its prevalence floor but only modestly.

Query (positives)	GeoRSCLIP	CLIP ViT-L/14	RemoteCLIP
new buildings/structures (809)	0.698	0.652	0.702
a new road or street (460)	0.418	0.377	0.372
bare ground or land cleared (401)	0.355	0.372	0.362
trees appeared/cleared (370)	0.341	0.313	0.334
low vegetation/grassland (311)	0.275	0.280	0.273
a water body (167)	0.168	0.152	0.159
a playground/sports field (33)	0.085	0.035	0.164
macro mAP (zero-shot)	0.334	0.312	0.338
macro mAP (naive)	0.453	0.430	0.441

Across seven change types every query clears its prevalence floor - broader recovery than DEN, where most queries sat at random - but only modestly: the zero-shot macro (≈ 0.33) is $\approx +0.04$ over the 0.297 floor, buildings (≈ 0.70) dominate, and water/playground are weak. As on QFabric, naive $>$ zero-shot (macro 0.45 vs 0.33): end-state appearance carries more of these land-cover categories than the temporal Δ . SECOND-CC thus extends the project’s central law to a wide open vocabulary:

recovery scales with the visual salience and prevalence of the change type, and is weak in absolute terms with frozen CLIP-variant encoders.

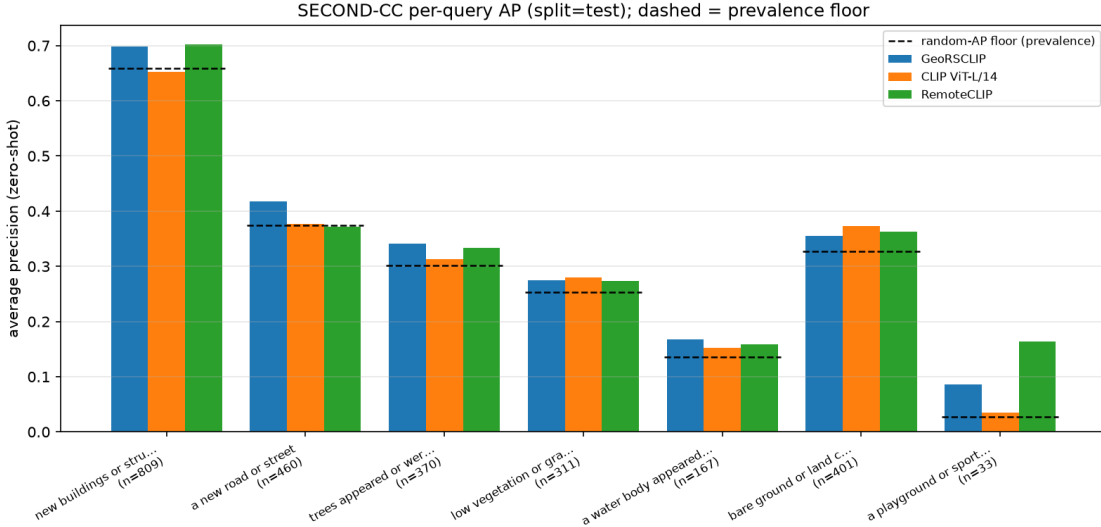


Figure 11: SECOND-CC per-query zero-shot AP by encoder (test split, RGB), with each query’s prevalence floor (dashed). Across all seven change types every query clears its floor - broader recovery than DEN - but only modestly, with buildings dominating and water/playground weakest.

Quantitative change localisation. The change heatmap is scored on the LEVIR-MCI and SECOND-CC pixel masks. For each masked query the query-conditioned heatmap (per-patch Δ -similarity $\cos(t, P_{2p}) - \cos(t, P_{1p})$, the same signal as the patch scorer of Section 5) is scored by pointing-game [24] (is the highest- Δ patch a true change patch?) and patch-AP, over pairs that contain the change with at least one positive patch at grid resolution. Each encoder is compared only to its own random-patch floor, since the patch grid differs by encoder (GeoRSCLIP 7×7 , the ViT-L encoders 16×16).

Table 15: LEVIR-MCI change localisation (test split; pointing-game and patch-AP against each encoder’s random-patch floor). Lift is pointing minus floor.

Encoder (grid)	Class	pointing	floor	lift
GeoRSCLIP (7×7)	road	0.416	0.334	+0.082
GeoRSCLIP (7×7)	building	0.351	0.378	-0.027
RemoteCLIP (16×16)	road	0.312	0.208	+0.104
RemoteCLIP (16×16)	building	0.257	0.238	+0.019
CLIP ViT-L/14 (16×16)	road	0.151	0.208	-0.057
CLIP ViT-L/14 (16×16)	building	0.101	0.238	-0.137

The heatmap is a weak localiser. Only road change is localised above chance, and only by the RS-pre-trained encoders (RemoteCLIP +0.10, GeoRSCLIP +0.08 pointing-game lift); building change is not localised above its floor by any encoder, and generic CLIP ViT-L *anti-localises* (-0.14 on building). On the SECOND-CC semantic masks the verdict matches - every reliably-sampled class sits within ± 0.04 of its random-patch floor (the lone large value, playground +0.30 on GeoRSCLIP, rests on 22 pairs and does not replicate). This is the spatial counterpart of the retrieval result: frozen CLIP-variant

embeddings recover change weakly, and localising it - a harder task than retrieving it - is weaker still, with only RS pre-training buying a modest road signal. The overlay stays useful for a human in the loop, but it is not a quantitatively reliable detector.

The salience law. Pooled across the open-vocabulary datasets (Figure 12), the recovered signal tracks change salience: weak on DEN’s subtle spectral transitions, dataset-dependent on QFabric, strong on LEVIR-CC’s salient building/road change (per-query AP $\approx 0.6\text{--}0.8$) yet near-random on its subtle vegetation and sparse water, and above every SECOND-CC prevalence floor but only modestly. This echoes — by analogy rather than shared mechanism — the classical visual-saliency observation [31] that salient stimuli dominate perception: here the driver is text–image embedding alignment, not bottom-up attention, yet salience is the single empirical regularity the project finds in every dataset.

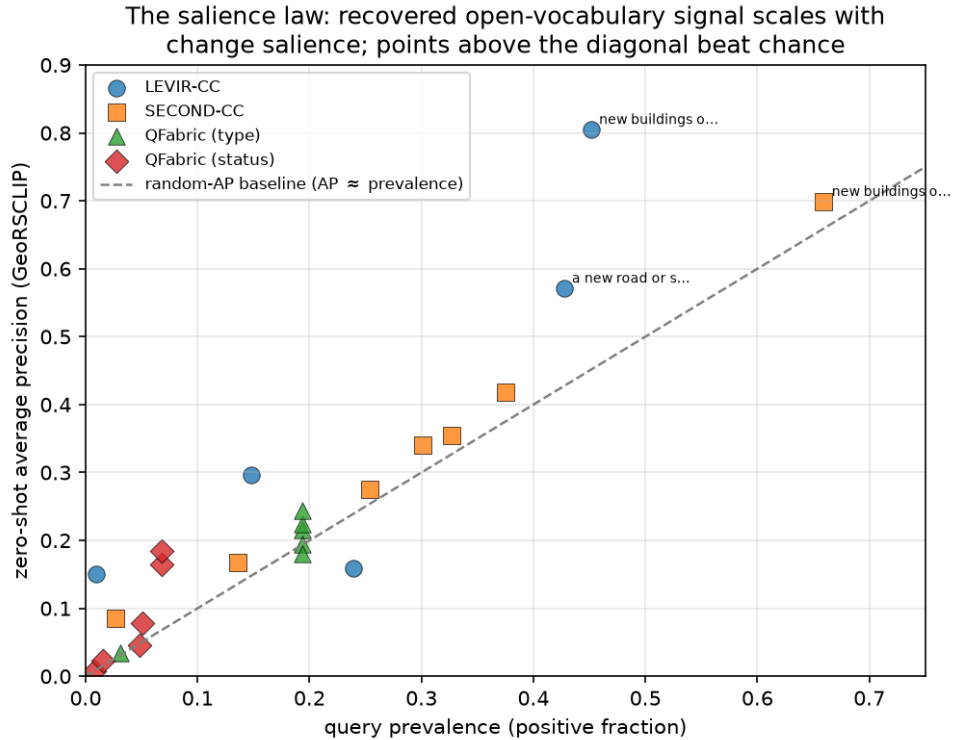


Figure 12: The salience law, pooled across the open-vocabulary datasets: each point is a per-query zero-shot AP (GeoRSCLIP) against that query’s prevalence, where a random ranker scores $AP \approx prevalence$ (dashed diagonal). Salient, well-described construction change (new buildings, roads) sits far above chance; subtle, sparse, or rare transitions (water, vegetation, status changes) collapse onto the diagonal - the single law this project finds in every dataset.

Qualitative retrieval: salient vs subtle change (georsclip, levir_mci)
 Actual top-1 zero-shot pair per query [Before | After | change heatmap]; green = relevant, red = not (query label predicate).
 The change heatmap is a weak localiser (report Sec. 8.5), shown for transparency.

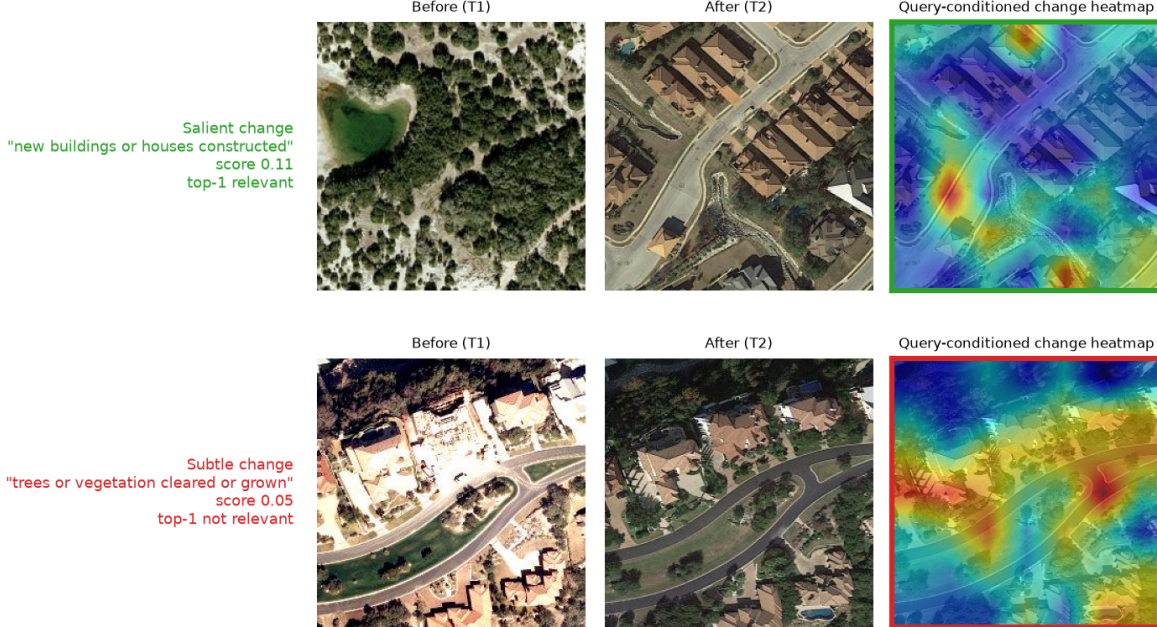


Figure 13: The salience law in a single view (LEVIR-CC, GeoRSCLIP, zero-shot). For a salient query (new buildings) and a subtle one (vegetation), the panels show the engine’s actual top-1 retrieved pair — Before, After, and the query-conditioned change heatmap — with a green border when that top-1 satisfies the query’s label predicate and red otherwise. The salient query retrieves a genuine construction event at rank 1, whereas the subtle query’s top-1 is a false positive, mirroring the aggregate per-query AP of Figure 12. Consistent with the localisation analysis (Table 15), the change heatmap is only a weak localiser and is shown for transparency, not as a localisation result; the absolute Δ -similarity scores (0.11 and 0.05) stay low, as expected under the frozen-VLM ceiling. Nothing is cherry-picked — the figure regenerates deterministically from cached embeddings via `scripts/make_qualitative_figure.py`.

9 Resources and operational metrics

All work fits the low-compute budget of the brief: a single laptop GPU, frozen backbones, and cached embeddings. Tables 16–19 summarise hardware, model sizes, disk footprint and timings.

Table 16: Hardware and software.

Item	Spec
GPU	NVIDIA GeForce RTX 4060 Laptop, ~8 GB VRAM
Runtime	torch 2.10.0 + CUDA (cu130); CPU fallback supported
OS / Python	Windows 11 / Python 3.12.10
Optional	free Kaggle / Colab GPU for heavier sweeps

Table 17: Model sizes. The adapter is <0.2% of the backbone - the PEFT premise.

Model	Weights on disk	Params (approx)	Dim
CLIP ViT-L/14 (HF)	≈1.71 GB	≈427 M	768
GeoRSCLIP (ViT-B/32 + RS5M ckpt)	605 MB ckpt	≈151 M	512
RemoteCLIP (ViT-L/14 + ckpt)	≈1.7 GB ckpt	≈428 M	768
PEFT adapter (trainable part)	2.1–2.9 MB	725,504 (768) / 528,128 (512)	-

Table 18: Disk footprint.

Component	Size
DEN archive (removable after extract)	7.09 GB
DEN extracted	9.03 GB
CLIP weights cache (in-repo, gitignored)	≈1.6 GB
HF hub cache (in-repo, gitignored)	≈2.2 GB
Embedding caches (per encoder, 605 pairs)	CLIP 3.76 MB, GeoRSCLIP 2.52 MB
Trained adapters	2.1–2.9 MB each
Total	≈19.5 GB (≈12.5 GB after deleting the archive)

Table 19: Timings (RTX 4060, dedicated timed pass).

Operation	Time
Retrieval scoring - numpy, 605 pairs, excl. text encode	0.269 ms/query
End-to-end query - CLIP text forward + scoring, 605 pairs	10.5 ms
Embedding precompute - CLIP L/14, 1024 ² → 224, GPU	68 ms/tile (≈82 s, one-time)
PEFT training - 605 samples, 40 epochs, adapter only, GPU	≈29 s
Fast test suite - mock encoders, CPU	≈65 s

10 Reproducibility

Seeds are fixed; embeddings and adapters are cached and keyed by (dataset, encoder, split, colour mode), with LoRA caches adding a `_lora` tag and cache invalidation on a pair-set change. The synthetic fixture is regenerated automatically by the test suite. The commands below are the reproducibility recipe used to produce the numbers in this report (a step-by-step guide is in `README.md`).

```

pip install -e .
python -m scripts.download_den --dest data/DynamicEarthNet # ~7 GB, one-time

# In-distribution: train on train split, evaluate on train/val/test
python -m scripts.run_pipeline --root data/DynamicEarthNet --encoder clip_vitl14 \
  --train-split train --eval-splits train val test --epochs 40
python -m scripts.run_pipeline --root data/DynamicEarthNet --encoder georsclip \
  --train-split train --eval-splits train val test --epochs 40
python -m scripts.run_pipeline --root data/DynamicEarthNet --encoder remotecclip \
  --train-split train --eval-splits train val test --epochs 40

# Best generalising config: GeoRSCLIP + NRG zero-shot (no PEFT training)
python -m scripts.run_pipeline --root data/DynamicEarthNet --encoder georsclip \
  --color-mode nrg --eval-splits train val test --skip-train

```

```

# NDVI / NRG colour-mode ablation (zero-shot only)
python -m scripts.run_pipeline --root data/DynamicEarthNet --encoder clip_vitl14 \
    --color-mode ndvi --eval-splits train val test --skip-train
python -m scripts.run_pipeline --root data/DynamicEarthNet --encoder georsclip \
    --color-mode ndvi --eval-splits train val test --skip-train
python -m scripts.run_pipeline --root data/DynamicEarthNet --encoder remoteclip \
    --color-mode nrg --eval-splits train val test --skip-train

# LoRA adapter on visual encoder (GeoRSCLIP + NRG)
python -m scripts.run_pipeline --root data/DynamicEarthNet --encoder georsclip \
    --color-mode nrg --skip-train \
    --lora --lora-epochs 20 --lora-rank 4 --lora-alpha 8 \
    --eval-splits train val test

# Gradio UI
python -m src.app --root data/DynamicEarthNet --encoder clip_vitl14 --split train

# Fast suite (deterministic, no network)
pytest -q --ignore=tests/test_text_encoder.py

```

The figures in this report are regenerated from cached embeddings (no GPU training) via `scripts/export_results.py` (writes one JSON per run + `macro_summary.csv`) followed by `scripts/make_figures.py` and `scripts/make_comparison_figure.py`. The main sources of run-to-run variance are GPU non-determinism in the encoders and the random partitions of the cross-validation; the cross-validation means and standard deviations reported throughout quantify the latter, and seeds are fixed for the former.

Code, data, and demonstration availability. The full source, the preprocessed-data download recipe, the trained adapters, and the committed benchmark results are available in the public repository at https://github.com/Panagiotis427/MSc_GBDA-0V_Temporal_Change_Retrieval. The Gradio search engine is deployed and publicly usable, over a bundled synthetic corpus, at https://huggingface.co/spaces/panagiotis427/Open_Vocabulary_Temporal_Change_Retrieval; five short screen-recorded walkthroughs of the interface, in the repository’s `demos/` directory, show the engine in use end to end.

11 Limitations and future work

The limitations group by how much they bound the conclusions. The first tier concerns credibility: all mAP figures use LULC-derived pseudo-labels rather than human-annotated query relevance, so a true information-retrieval benchmark would require collecting human relevance judgements, the single change that would most strengthen the evaluation. The second tier concerns the method: supervision is weak (captions are derived from dominant-class label flips), so LLM-generated or human-written change captions are a natural improvement; both the ProjectionHead and LoRA adapters overfit train-AOI statistics, so multi-AOI held-out training or domain-randomised augmentation might close the out-of-distribution gap; and Sentinel-1 [32]/2 [19] fusion is unimplemented, though `aoi_metadata.json` confirms 51/75 AOIs carry full SAR (S1) coverage and feeding SAR Δ -features would be a direct extension of the NRG pattern. The third tier concerns scope: crop-precise polygon grounding for finer localisation remains open — the QFabric pentatemporal five-date imagery with polygon change-masks is the natural vehicle for it, but that source is currently access-gated; fMoW is wired only at the protocol level with no concrete loader; and a learned query-to-patch attention head is unexplored because every trained head on this data memorises rather than generalises.

12 Conclusion

The system fulfils the GBDA Lab Project’s brief: an open-vocabulary, frozen-backbone change search engine with a Gradio interface, a label-grounded Recall@ K /mAP benchmark, seasonal-vs-permanent error analysis, temporal pinpointing of the change within each AOI timeline (Appendix B), and a clear zero-shot vs PEFT comparison across three CLIP-variant encoders on Dynamic EarthNet, reported under a strict statistical protocol. Four findings emerge. First, a learned adapter shows no out-of-distribution gain: the large in-distribution PEFT lift ($0.043 \rightarrow 0.420$, up to 0.999 on QFabric) is the adapter scored on its own training pairs (memorisation), and under the headline GeoRSCLIP+NRG configuration leakage-free cross-validated PEFT (0.196 ± 0.049) merely overlaps frozen zero-shot (0.139 ± 0.024) within fold variance, so low-compute fine-tuning is not shown to help retrieval out of distribution. Second, out of distribution the frozen zero-shot wins, but weakly: the GeoRSCLIP+NRG estimate is 0.100 ± 0.139 under 5-fold AOI cross-validation (the single-split 0.426 is a high-variance fold), so RS pre-training and NIR false-colour help, but the absolute signal is small. Third, the recoverable signal is a matter of evaluation and method, not bad data: pixel-fraction relevance makes 9/10 change-types evaluable and a localised patch-level scorer lifts cross-validated mAP to 0.193, making construction change (new buildings, urban expansion) retrievable above chance, with only snow genuinely absent from the data. Fourth, LoRA does not close the gap either (every rank/epoch overfits to test ≈ 0.07), consistent with the adapters-only-memorise finding.

The deliverable’s value is the motivated, honestly-audited comparison, not a single best number: spectral physics and spatial locality carry the weak signal that exists, global embeddings and learned visual priors dilute or memorise it, and small single-split scores require cross-validation before they estimate generalisation. Across all four datasets the recovered signal tracks change salience (Figure 12): weak on DEN’s subtle spectral transitions, dataset-dependent on QFabric, strong on LEVIR-CC’s salient building/road change (per-query AP ≈ 0.6 – 0.8) yet near-random on its subtle vegetation and sparse water, and - on SECOND-CC’s seven-query open vocabulary - above every prevalence floor but only modestly (zero-shot macro ≈ 0.33). Localising the change (LEVIR-MCI/SECOND-CC masks) is weaker still: the heatmap clears its random-patch floor by at most ≈ 0.04 – 0.10 . The same regularity governs the temporal axis (Appendix B): sharp, visually distinct events such as snow melt are pinpointed in time above chance, whereas gradual transitions such as deforestation collapse toward the random floor – so change salience sets the recoverable signal in time as well as in space and vocabulary.

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Appendix A Native 3 m raster source: data-source fidelity

The primary evaluation uses the ~ 7 GB preprocessed RGB/NIR-JPEG DEN subset in preference to the ~ 525 GB raw mirror. To confirm that this pragmatic choice does not cost retrieval performance - i.e. that the conclusions survive at full radiometric depth - the adapter was also run directly on the native Planet-Fusion [5] surface-reflectance rasters ($1024 \times 1024 \times 4$ `int16`, Blue/Green/Red/NIR, 3 m), read in-memory from the per-zone `planet.<UTM>.zip` archives with no lossy re-encoding, yielding 23 labelled AOIs (552 change pairs, under that earlier pipeline’s denser pairing) spanning all seven DEN land-cover classes. The adapter (frozen CLIP ViT-L/14, **difference** change feature) was scored under the same held-out protocol as Section 8.1: a 5-seed stratified AOI-level hold-out (split on real cube id, stratified by dominant change class), reporting mean $\pm$ std across seeds. These particular figures were produced by an earlier iteration of this project’s codebase (the `cross_validate.py` pipeline over `train_v2.py` and `evaluate_v2.py`), which is not part of the committed source; the run is therefore a one-off that cannot be regenerated from the current tree. The per-seed fold values are archived verbatim in `results/cv_eval__clip_vitl14__rgb__planet3m.json`, and the reproducible native-3m counterpart — run through this repository’s own committed `cv_eval` harness — is reported in Table 21. The adapter table below is retained for the source-fidelity comparison it supports, read with that provenance caveat in mind.

Table 20: Native 3 m raster source (CLIP ViT-L/14, adapter, **difference**; 23 AOIs, 5-seed stratified AOI cross-validation). NIR here is the NIR-Red-Green false colour (matching `nrg` elsewhere). These figures come from a one-off run on the full-resolution Planet-Fusion rasters, external to the committed `results/` artifacts (which cover the ~ 7 GB JPEG subset); its generating pipeline is not committed, so these figures are not reproducible from the tree (per-seed values archived in the run’s JSON). Reported for the source-fidelity comparison only; reproducible counterpart in Table 21.

Configuration	colour	mAP (mean $\pm$ std)
Adapter	RGB	0.236 ± 0.030
Adapter	NIR (NRG)	0.253 ± 0.040
Adapter, class-weighted	RGB	0.265 ± 0.029
Adapter, 2×2 patch tiling	RGB	0.202 ± 0.022

Two readings follow, both consistent with the JPEG-subset audit. First, the native rasters land in the same band as the preprocessed subset under cross-validation (cf. the audited CV figures in Section 8.1): full `int16` radiometric depth and the intact NIR channel buy no material retrieval gain over the ~ 7 GB RGB/NIR-JPEG subset, so the data-source choice in Section 6 is vindicated - the bottleneck is the frozen web-pretrained features and the small labelled AOI set, not imagery compression. Second, no full-tile configuration is statistically separable from another: RGB (0.236 ± 0.030), NRG (0.253 ± 0.040) and per-class loss weighting (0.265 ± 0.029) all overlap within one standard deviation, and the $+0.017$ NRG-over-RGB margin sits well inside that noise - the same null result as the colour-mode ablation (Section 8.3), where the large single-split NRG lift is a high-variance fold. Only 2×2 patch tiling separates, and downward (0.202 ± 0.022): splitting the 1024-px tile discards the CLIP global spatial context. RGB at the native tile scale is the simplest configuration in the indistinguishable top group and remains the reference.

Cross-check under the repository’s own benchmark. The table above reports the fitted adapter under a 5-seed stratified AOI hold-out. As a second, independent check — this time zero-shot and under the disjoint 5-fold AOI `cv_eval` harness that produced the JPEG-subset headline (dominant-class relevance over the ten DEN queries; Section 8.1), via the registry-resolved `dynamic_earthnet_planet` loader — the native rasters were scored directly against the preprocessed subset. Both use the same dominant-class relevance metric, so the two rows are read as adapter-vs-zero-shot, not metric-vs-metric.

Table 21 reports the head-to-head. Under the identical encoder, metric and zero-shot setting the native 3 m source (macro mAP 0.130 ± 0.068) sits at or above the JPEG subset (0.076; 0.123 under fraction relevance), despite covering fewer AOIs. The comparison is not fully controlled — the corpora differ in AOI count (23 vs 75) and colour composite (native RGB vs JPEG NRG) — but for this corpus colour barely moves the score (georsclip: RGB 0.085 vs NRG 0.100), so the direction is robust: the native source is no worse, corroborating the no-loss conclusion of Section 6 under the repository’s own zero-shot cross-validated benchmark rather than the fitted hold-out above.¹ The two native-source adapter estimates also diverge sharply in stability — the external hold-out’s tight 0.236 ± 0.030 (Table 20) against the leakage-free `cv_eval` adapter’s 0.215 ± 0.306 — so the fitted-adapter level is not read as a gain over zero-shot: the instability under leakage-free folds is exactly the behaviour Section 8.2 attributes to adapter memorisation on this small labelled corpus, and the source-fidelity reading accordingly rests on the stable, reproducible zero-shot rows.

Table 21: Native 3 m rasters vs the preprocessed JPEG subset under the repository’s own `cv_eval` benchmark (CLIP ViT-L/14, 5-fold AOI cross-validation, zero-shot, dominant-class relevance unless noted).

Source	colour	macro mAP (mean $\pm$ std)
Native 3 m raster	RGB	0.130 ± 0.068
JPEG subset	NRG	0.076
JPEG subset (fraction rel.)	NRG	0.123

Controlled compression and resolution ablation. The cross-checks above leave two confounds (AOI count and colour composite). To isolate image fidelity itself, everything is held fixed — the same 23 AOIs, 253 pairs, RGB composite, encoder, zero-shot scoring and 5 AOI folds — and only the degradation applied to each tile before encoding is varied. The native lossless raster is the upper bound; every other row is the identical corpus pushed through a lossy round-trip in memory. Relevance labels (from the lossless masks) and the text-query vectors are computed once and shared across all rows, so the only thing that changes between rows is the image. Two degradation families are swept: JPEG [33] quality $q \in \{95, 75, 50, 25, 10\}$ (radiometric / compression loss at full resolution), and bicubic downsampling to $\{512, 256, 128, 64\}$ px followed by JPEG $q=75$ (the resolution loss the preprocessed subset’s resize step would introduce). Table 22 and Figure 14 report CLIP ViT-L/14.

¹The leakage-free k -fold PEFT estimate on the native rasters is 0.215 ± 0.306 ; the large variance (one fold at 0.75, the rest 0.03–0.19, driven by few positives per small fold) makes the zero-shot figure the reliable point of comparison.

Table 22: Controlled degradation ablation on the native 3 m rasters (CLIP ViT-L/14, RGB, zero-shot, 253 pairs / 23 AOIs, 5-fold AOI cross-validation, seed 42). Identical corpus throughout; only the per-tile degradation varies. Every row overlaps native within one fold standard deviation.

Degradation	macro mAP (mean $\pm$ std)	Δ vs native
Native 3 m (lossless, 1024 px)	0.130 $\pm$ 0.068	—
JPEG $q=95$	0.115 $\pm$ 0.065	−0.015
JPEG $q=75$	0.094 $\pm$ 0.064	−0.036
JPEG $q=50$	0.128 $\pm$ 0.078	−0.001
JPEG $q=25$	0.095 $\pm$ 0.065	−0.034
JPEG $q=10$	0.138 $\pm$ 0.108	+0.009
Downsample 512 px (+ JPEG $q75$)	0.088 $\pm$ 0.054	−0.042
Downsample 256 px	0.154 $\pm$ 0.150	+0.025
Downsample 128 px	0.146 $\pm$ 0.166	+0.016
Downsample 64 px	0.216 $\pm$ 0.213	+0.086

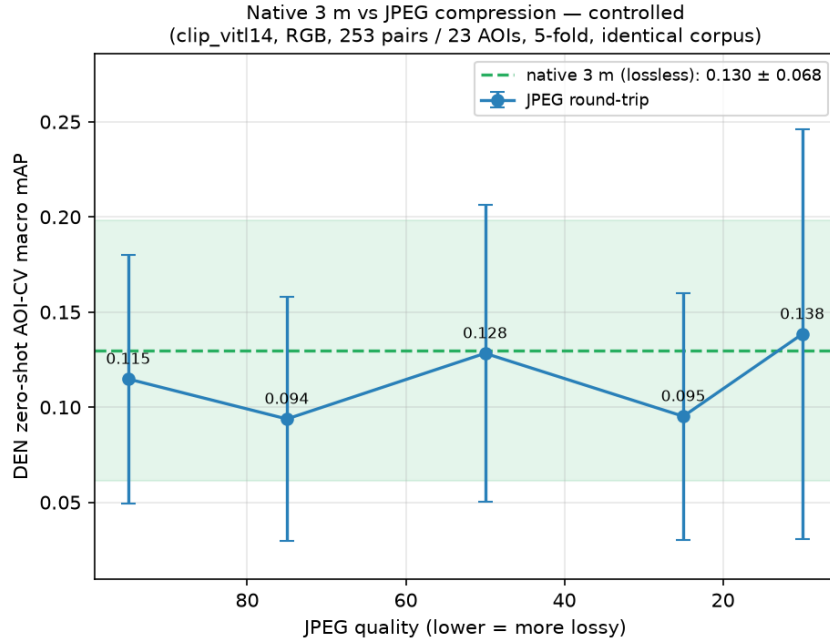


Figure 14: Zero-shot AOI-CV macro mAP versus JPEG quality on the identical native corpus (CLIP ViT-L/14, RGB). The dashed line and shaded band are the native lossless mean $\pm$ fold std. Every JPEG point sits inside the band, and the curve is non-monotonic — the most-compressed setting ($q=10$) scores above the lightly-compressed ones — so the variation is fold sampling, not a compression effect.

The reading is unambiguous: image fidelity is not the bottleneck. The largest gap from native across the entire JPEG sweep is 0.036 mAP — about half of one fold’s standard deviation (~ 0.065) — and the deltas alternate sign with quality, so they are fold-sampling noise rather than a degradation signal. The resolution sweep is, if anything, more decisive: collapsing the tile to 64 px (a $16\times$ linear reduction) does not lower retrieval and nominally raises it, because the frozen encoder resizes every input to its own 224 px window and the change signal that survives in the global embedding is coarse —

high-frequency spatial detail is simply not what the web-pretrained features exploit for these queries.² This is direct evidence for the report’s central thesis (Section 8.1 and Appendix A): the ceiling is the frozen web-pretrained representation and the small labelled-AOI set, not imagery compression or resolution. It also retroactively vindicates the Section 6 decision to work from the ~ 7 GB preprocessed subset rather than the ~ 525 GB native mirror.

The result is not an artefact of the generic web-pretrained encoder. Repeating the identical ablation with the RS-pretrained GeoRSCLIP (ViT-B/32) gives the same null (Table 23): the largest deviation from its native baseline (0.085 ± 0.059) is 0.027 mAP, again inside one fold standard deviation, and every JPEG and downsample row overlaps native. Tellingly, the two encoders disagree on the sign of the small downsampling deltas — CLIP nominally rises toward 64 px, GeoRSCLIP nominally falls — which is exactly what fold-sampling noise, and not a real fidelity effect, would produce. Both the domain-agnostic and the RS-domain backbone are insensitive to compression and resolution on this task.

Table 23: Confirmation under GeoRSCLIP (ViT-B/32, RGB, zero-shot, same 253 pairs / 23 AOIs, 5-fold AOI cross-validation, seed 42). Same null as Table 22: every row within one fold standard deviation of native.

Degradation	macro mAP (mean $\pm$ std)	Δ vs native
Native 3 m (lossless, 1024 px)	0.085 ± 0.059	—
JPEG $q=95$	0.083 ± 0.054	-0.002
JPEG $q=75$	0.095 ± 0.075	$+0.010$
JPEG $q=50$	0.091 ± 0.058	$+0.006$
JPEG $q=25$	0.091 ± 0.073	$+0.006$
JPEG $q=10$	0.104 ± 0.079	$+0.019$
Downsample 512 px (+ JPEG $q75$)	0.102 ± 0.082	$+0.017$
Downsample 256 px	0.076 ± 0.059	-0.010
Downsample 128 px	0.074 ± 0.075	-0.011
Downsample 64 px	0.058 ± 0.035	-0.027

²The honest caveat is that fold variance grows under aggressive downsampling (std up to 0.21 at 64 px, driven by few positives per small fold), so the claim is only that no degradation is detectable at this corpus size, not exact equivalence.

Appendix B Temporal pinpointing: when does the change occur?

The project brief singles out Dynamic EarthNet for its ability to “pinpoint exact time-steps of change” and asks evaluation to verify that the model identifies “the specific temporal window where the change was most prominent compared to stable time-steps”. The retrieval metrics so far answer which pair shows a change (ranking across AOIs); this is the orthogonal question of when a change happens (ranking time-steps within one AOI’s timeline), and it is uniquely accessible through the native loader’s monthly raster stack. The loader is built with monthly pairing so each AOI’s consecutive pairs form its 24-month timeline; the per-step change score is the same zero-shot Δ -similarity used throughout ($\cos(q, f_t) - \cos(q, f_{t-1})$); a step is relevant when the query predicate fires on its monthly transition label. Within each AOI the steps are ranked and the ranking is scored by average precision (a per-AOI temporal AP), and the peak-hit rate is reported — whether the single highest-scoring step is a true transition, at exact and ± 1 -month tolerance. Significance uses a permutation test that shuffles relevance within each AOI (preserving its step count) plus BH-FDR.

Table 24: Temporal pinpointing on native 3 m DEN (CLIP ViT-L/14, RGB, zero-shot): ranking the monthly steps within each AOI timeline. “peak ± 1 ” is the fraction of AOIs whose top-scored step is within one month of a true transition. Macro temporal mAP 0.309 vs a 0.207 permutation floor.

Query	temporal mAP	rand	peak ± 1 mo	perm p	AOIs
seasonal snow melting away	0.586	0.200	1.00	0.025	2
bare soil or land cleared	0.324	0.214	0.57	0.067	7
forest loss	0.273	0.193	0.40	0.160	5
new water body or flooding	0.221	0.246	1.00	0.409	1
deforestation, forest $\rightarrow$ soil	0.142	0.183	0.20	0.617	5
macro	0.309	0.207	0.63	—	—

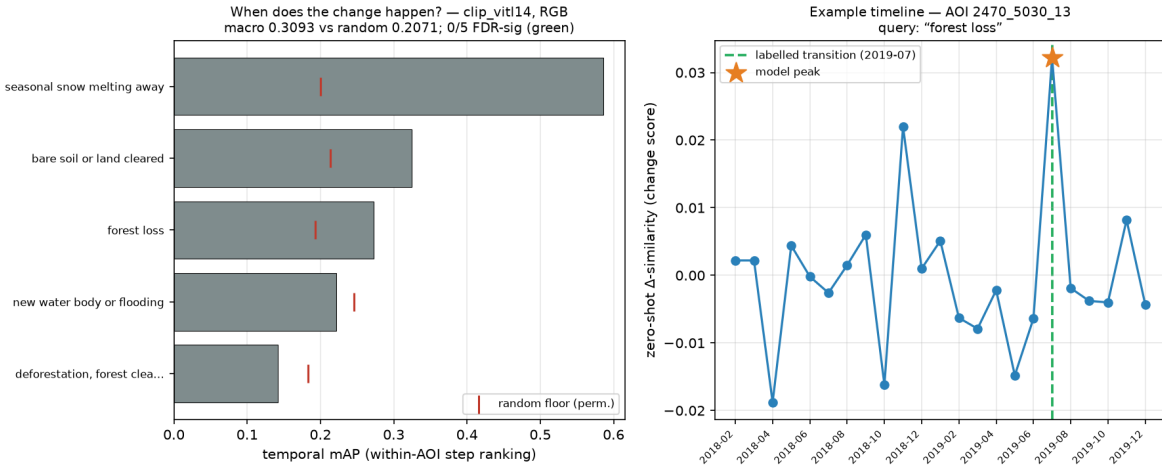


Figure 15: Left: per-query temporal mAP (within-AOI step ranking) against the permutation random floor (red ticks). Right: an example AOI timeline — the zero-shot change score for “forest loss” spikes at the labelled transition month (2019-07, dashed green), where the model’s peak (orange star) also lands.

The system does locate change in time above chance: macro temporal mAP 0.309 against a 0.207

random floor, and within a ± 1 -month tolerance the model’s peak step lands on a true transition 63% of the time (Table 24, Figure 15). The signal is strongest for temporally sharp events — snow melt (0.586, $p = 0.025$) and abrupt soil clearing (0.324) — and weak for gradual or ambiguous ones (deforestation 0.142, below its floor). As with spatial localization (Section 8.5) the effect is encoder-dependent: repeating the study with GeoRSCLIP (ViT-B/32) gives macro 0.146 vs the same 0.207 floor — at or below chance — so the larger general-purpose ViT-L/14 pinpoints change in time where the smaller RS-pretrained backbone does not. The honest limit is statistical power: only 1–7 AOIs carry positives per query, so although the aggregate clears the floor and snow melt is individually significant at $p = 0.025$, no query survives BH-FDR — the same small-labelled-AOI ceiling documented throughout. Pinpointing when a change occurs is feasible for salient, abrupt transitions on a capable frozen encoder, but, like pinpointing where, it is harder than retrieving whether the change is present.

Appendix C Post-retrieval re-ranking

The Gradio app exposes two optional, per-query re-ranking toggles. Both are evaluated on the held-out test split (110 pairs, 3 queries with positives) using GeoRSCLIP + NRG zero-shot (Table 25).

Table 25: Re-ranking on the test split (GeoRSCLIP + NRG, zero_shot). Both strategies reduce retrieval quality. Best per column in bold.

Strategy	R@1	R@3	R@5	R@10	mAP
baseline (no rerank)	0.133	0.333	0.400	0.467	0.426
diversity	0.133	0.267	0.267	0.267	0.344
coherence	0.133	0.200	0.267	0.467	0.339

Both strategies reduce retrieval quality: diversity deduplicates locations, pushing relevant pairs from the same AOI out of the top window; coherence clusters geographically near the top-1 result, but relevant pairs are globally distributed. Neither optimises semantic relevance - they trade mAP (-0.082 diversity, -0.087 coherence) for display properties (result variety and spatial coherence), and are exposed as optional toggles alongside a continental geographic filter driven by `aoi_metadata.json`.